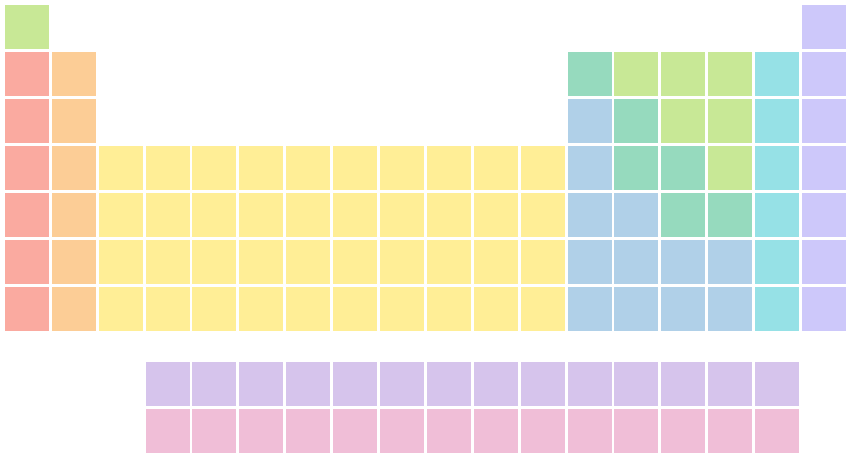
PERIODIC TABLE & PERIODICITY

Formula • Derivation • Conditions • Master Atlas

Complete derivation companion for JEE Main & JEE Advanced (Inorganic / Physical Chemistry)



Contents of this atlas. Modern long-form table with all 118 elements · IUPAC nomenclature · s-, p-, d-, f-block classification · metals, non-metals, metalloids · magic numbers and element location · screening and Z_{eff} (Slater) · periodicity · atomic radius · ionisation enthalpy · electron gain enthalpy · electronegativity · diagonal relationship.

Reference source: ALLEN JEE Inorganic Chemistry (Nurture Course) – Periodic Table.
Design reference: Thermodynamics & Thermochemistry Formula Book.

Conventions used throughout. ΔH_{eg} is the electron gain enthalpy (negative = energy released); electron affinity EA $= -\Delta H_{\text{eg}}$ for the first electron. Ionisation enthalpy $\Delta_{\text{i}}H > 0$ always. Radii are in picometres ($1\ \text{pm} = 10^{-12}\ \text{m} = 0.01\ \text{Å}$). χ denotes Pauling electronegativity. All boxed results are final, exam-ready formulas; every box is preceded by its derivation or justification and followed by the conditions under which it applies.

Contents

I Foundations . . . 3
1 Introduction 3
1.1 Early attempts at classification . . . 3
1.2 Moseley's discovery and the Modern Periodic Law . . . 4
2 Long Form (Present Form) of the Modern Periodic Table 4
2.1 Groups and families . . . 5
2.2 Description of periods . . . 5
2.3 Block structure of the table . . . 6
2.4 The modern periodic table of 118 elements . . . 6
3 Nomenclature of Elements (IUPAC Systematic Names) 8
II Classification of Elements . . . 8
4 Classification by Blocks 9
4.1 s-Block elements . . . 9
4.2 p-Block elements . . . 9
4.3 d-Block elements . . . 9
4.4 f-Block elements . . . 10
5 Types of Elements 10
5.1 Classification based on electronic configuration . . . 10
5.2 Inert (noble) gases . . . 11
5.3 Representative elements . . . 11
5.4 Transition elements . . . 11
5.5 Inner transition elements . . . 11
5.6 Metals . . . 11
5.7 Non-metals . . . 11
5.8 Metalloids (semi-metals) . . . 12
III Locating an Element in the Table . . . 12
6 Predicting the Atomic Number of Successive Members of a Group 12
6.1 Magic numbers . . . 12
7 Determination of Period, Block and Group of an Element 13
7.1 Step-by-step procedure from the atomic number . . . 14
7.2 Illustrations . . . 14
IV Shielding and Periodicity . . . 15
8 Screening Effect and Effective Nuclear Charge 15
8.1 Concept . . . 15
8.2 Calculation of σ: Slater's rules . . . 16
8.3 Periodic variation of Z_{eff} . . . 17
9 Periodicity 18
9.1 Meaning and causes . . . 18
9.2 Periodic properties . . . 18

9.3 Valency . . . 19

V Periodic Properties . . . 19

10 Atomic Radius 19

10.1 Definition and difficulty . . . 19
10.2 How to calculate the atomic radius . . . 20
10.3 Types of atomic radius . . . 22
10.4 Ionic radii . . . 22
10.5 Factors affecting atomic radius . . . 23
10.6 Periodic trend . . . 23

11 Ionisation Enthalpy 25

11.1 How to calculate ionisation enthalpy . . . 25
11.2 Successive ionisation enthalpies . . . 26
11.3 Factors affecting ionisation enthalpy . . . 27
11.4 Periodic trend . . . 29
11.5 Applications of ionisation enthalpy . . . 30

12 Electron Gain Enthalpy (Electron Affinity) 31

12.1 How to calculate electron gain enthalpy . . . 32
12.2 Successive electron gain enthalpies . . . 32
12.3 Factors affecting electron gain enthalpy . . . 33
12.4 Periodic variation of electron gain enthalpy . . . 33

13 Electronegativity 34

13.1 How to calculate electronegativity . . . 35
13.2 Typical values (Pauling scale) . . . 36
13.3 Difference between electronegativity and electron affinity . . . 36
13.4 Factors affecting electronegativity . . . 37
13.5 Periodic variation . . . 37
13.6 Applications of electronegativity . . . 38

14 Diagonal Relationship 39

14.1 Lithium resembles magnesium . . . 39
14.2 Beryllium resembles aluminium . . . 40
14.3 Boron resembles silicon . . . 40

VI Exam Tools . . . 40

15 Master Formula Atlas 40

16 Keyword Decision Tree 41

17 Symbol and Sign Glossary 42

Part I — Foundations

1 Introduction

The periodic table is the organising principle of chemistry. By arranging the elements into groups and periods we can study them and their compounds comparatively, instead of memorising the properties of more than a hundred elements one by one, and we can predict the trend of properties such as ionisation enthalpy, electron gain enthalpy, electronegativity, atomic size and chemical reactivity. Every topic in this book rests on one idea: *when elements are arranged in order of increasing atomic number, the same outer-shell electronic configuration recurs at regular intervals, and therefore similar properties recur.*

1.1 Early attempts at classification

Historical classifications were proposed by Prout (hypothesis that atomic masses are integral multiples of that of hydrogen), Döbereiner (triads), Newlands (law of octaves), Lothar Meyer (atomic-volume curve) and Mendeleev (periodic law based on atomic mass). The detailed history is in NCERT; the examinable essentials are collected below.

Scientist	Basis of classification	Merit	Limitation
Döbereiner (1829)	Triads: three similar elements, mass of the middle ≈ mean of the other two	First recognition of a numerical relation between mass and properties	Only a few triads could be found
Newlands (1864)	Law of octaves: every 8th element resembles the 1st	Showed periodic recurrence	Failed beyond Ca; no place for undiscovered elements
Lothar Meyer (1869)	Plot of atomic volume vs. atomic mass; alkali metals at the peaks	Showed periodicity of a physical property	Based on atomic mass
Mendeleev (1869)	Properties are a periodic function of atomic *mass*; 63 elements known, noble gases not yet discovered	Left gaps and predicted eka-aluminium (Ga), eka-silicon (Ge), eka-boron (Sc)	Anomalous pairs (Te–I, Ar–K, Co–Ni); no place for isotopes, lanthanoids; position of H uncertain

Döbereiner's triads. If three elements form a triad then the atomic mass of the middle element is the arithmetic mean of the other two.

Formula 1.1 Döbereiner Triad Rule

$$A_r(\text{middle}) \approx \frac{A_r(\text{first}) + A_r(\text{third})}{2}$$

Triad	Masses	Mean of extremes	Actual middle	Deviation
Li, Na, K	6.9, 23.0, 39.0	$\frac{6.9+39.0}{2} = 22.95$	23.0	0.05
Ca, Sr, Ba	40.1, 87.6, 137.3	$\frac{40.1+137.3}{2} = 88.7$	87.6	1.1
Cl, Br, I	35.5, 79.9, 126.9	$\frac{35.5+126.9}{2} = 81.2$	79.9	1.3

1.2 Moseley's discovery and the Modern Periodic Law

Mendeleev's anomalies (for example, tellurium, $A_r = 127.6$, placed *before* iodine, $A_r = 126.9$) vanish when elements are ordered by atomic number. In 1913 Moseley bombarded metals with electrons and measured the frequency ν of their characteristic X-rays.

Derivation. For the K_α line an electron falls from $n = 2$ to $n = 1$. The remaining $1s$ electron screens one unit of nuclear charge, so the electron experiences $(Z - 1)$. Using the Bohr–Rydberg expression with $Z \to Z - 1$:

$$\nu = R\,c\,(Z-1)^2\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = \frac{3}{4}\,R\,c\,(Z-1)^2 \quad \Longrightarrow \quad \sqrt{\nu} = \sqrt{\tfrac{3Rc}{4}}\,(Z-1)$$

Formula 1.2 Moseley's Law

$$\boxed{\sqrt{\nu} = a\,(Z-b)} \qquad \text{with } a = \sqrt{\tfrac{3Rc}{4}} \approx 4.97 \times 10^7 \text{ Hz}^{1/2}, \quad b = 1 \ (K_\alpha)$$

Conditions

A plot of $\sqrt{\nu}$ against Z is a straight line for each X-ray series, so the **atomic number** (nuclear charge) and not atomic mass is the fundamental property. The wavelength of the X-ray decreases (frequency increases) regularly from lighter to heavier element.

Modern Periodic Law

The physical and chemical properties of the elements are periodic functions of their atomic numbers.

2 Long Form (Present Form) of the Modern Periodic Table

The long form is also called the *Bohr–Bury–Rang–Werner* table. It is based on the Bohr–Bury electronic-configuration concept and the modern periodic law. It contains **7 periods** (horizontal rows) and **18 groups** (vertical columns), numbered 1 to 18 by IUPAC. Elements of the same group have the same number of electrons in the outermost shell and hence similar properties.

2.1 Groups and families

IUPAC	Old label	Family name	Valence configuration	Members
1	IA	Alkali metals	ns^1	Li, Na, K, Rb, Cs, Fr (H placed on top)
2	IIA	Alkaline-earth metals	ns^2	Be, Mg, Ca, Sr, Ba, Ra
3	IIIB	Scandium group	$(n-1)d^1ns^2$	Sc, Y, La, Ac (and the f-block)
4–7	IVB–VIIB	Ti, V, Cr, Mn groups	$(n-1)d^{2-5}ns^2$ (Cr: d^5s^1)	Ti Zr Hf Rf ⋯ Mn Tc Re Bh
8–10	VIII	Iron, cobalt, nickel triads (platinum metals below)	$(n-1)d^{6-8}ns^2$	Fe Ru Os; Co Rh Ir; Ni Pd Pt
11	IB	Coinage metals	$(n-1)d^{10}ns^1$	Cu, Ag, Au, Rg
12	IIB	Zinc group	$(n-1)d^{10}ns^2$	Zn, Cd, Hg, Cn
13	IIIA	Boron family	ns^2np^1	B, Al, Ga, In, Tl, Nh
14	IVA	Carbon family	ns^2np^2	C, Si, Ge, Sn, Pb, Fl
15	VA	Pnictogens (nitrogen family)	ns^2np^3	N, P, As, Sb, Bi, Mc
16	VIA	Chalcogens	ns^2np^4	O, S, Se, Te, Po, Lv
17	VIIA	Halogens	ns^2np^5	F, Cl, Br, I, At, Ts
18	0	Noble (inert) gases	ns^2np^6 ($1s^2$ for He)	He, Ne, Ar, Kr, Xe, Rn, Og

Placement of hydrogen and helium

Hydrogen ($1s^1$) resembles the alkali metals (loses one electron to form H^+, is a reducing agent) and also the halogens (needs one electron to complete its shell, forms H^- in saline hydrides, exists as H_2). It is conventionally placed at the top of group 1 but is a non-metal. Helium ($1s^2$) is an s-block element by configuration but is placed in group 18 because its shell is complete.

2.2 Description of periods

2.2.1 Why the periods have 2, 8, 8, 18, 18, 32, 32 elements

Sub-shells are filled in the order of increasing $(n+l)$; for equal $(n+l)$ the sub-shell of lower n fills first (Madelung or Aufbau rule). Every period begins with an ns sub-shell and ends with the np sub-shell (noble gas), so the length of a period equals twice the number of orbitals filled in it.

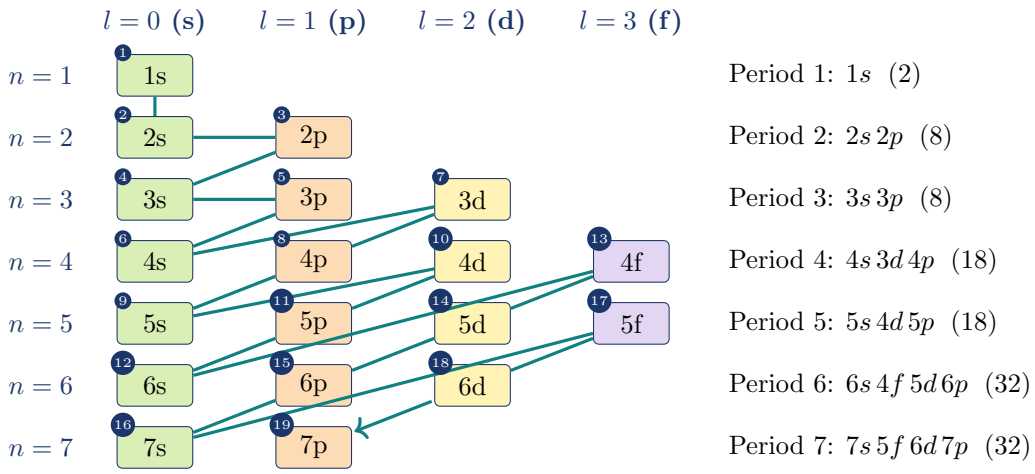


Figure 1. Order of filling of sub-shells (numbers give the sequence) and the periods they build.

Formula 2.1 Madelung Rule and Period Length

lower $(n+l)$ fills first; if equal, lower n

$$N_p = 2\left\lfloor \frac{p+2}{2} \right\rfloor^2$$

Conditions

N_p gives $2, 8, 8, 18, 18, 32, 32$ for $p = 1, \ldots, 7$ (floor function: round down). Check for $p = 6$: sub-shells $6s(1) + 4f(7) + 5d(5) + 6p(3) = 16$ orbitals, so $N_6 = 2 \times 16 = 32$. The numbers $2, 8, 8, 18, 18, 32, 32$ are called **magic numbers** of the table. Period lengths are *not* $2n^2$, because of the $(n+l)$ ordering.

Period	n	Sub-shells filled	Orbitals	Elements	Range (Z)	Name of period
1	1	$1s$	1	2	H–He (1–2)	Shortest
2	2	$2s, 2p$	4	8	Li–Ne (3–10)	Short
3	3	$3s, 3p$	4	8	Na–Ar (11–18)	Short
4	4	$4s, 3d, 4p$	9	18	K–Kr (19–36)	Long
5	5	$5s, 4d, 5p$	9	18	Rb–Xe (37–54)	Long
6	6	$6s, 4f, 5d, 6p$	16	32	Cs–Rn (55–86)	Longest
7	7	$7s, 5f, 6d, 7p$	16	32	Fr–Og (87–118)	Longest (now complete)

Update to older notes

Older books (and the ALLEN notes) list the seventh period as *incomplete* with 26 elements (up to $Z = 112$). After the discovery and naming of elements 113–118 (Nh, Fl, Mc, Lv, Ts, Og) the seventh period is complete with 32 elements and the table contains **118 elements**. Always read the date of the question source: an old-style question may still expect "26".

2.3 Block structure of the table

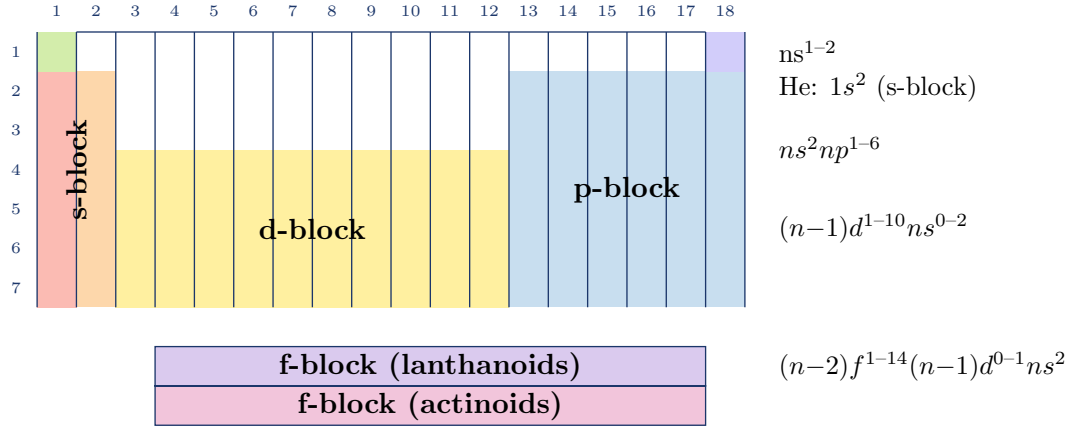
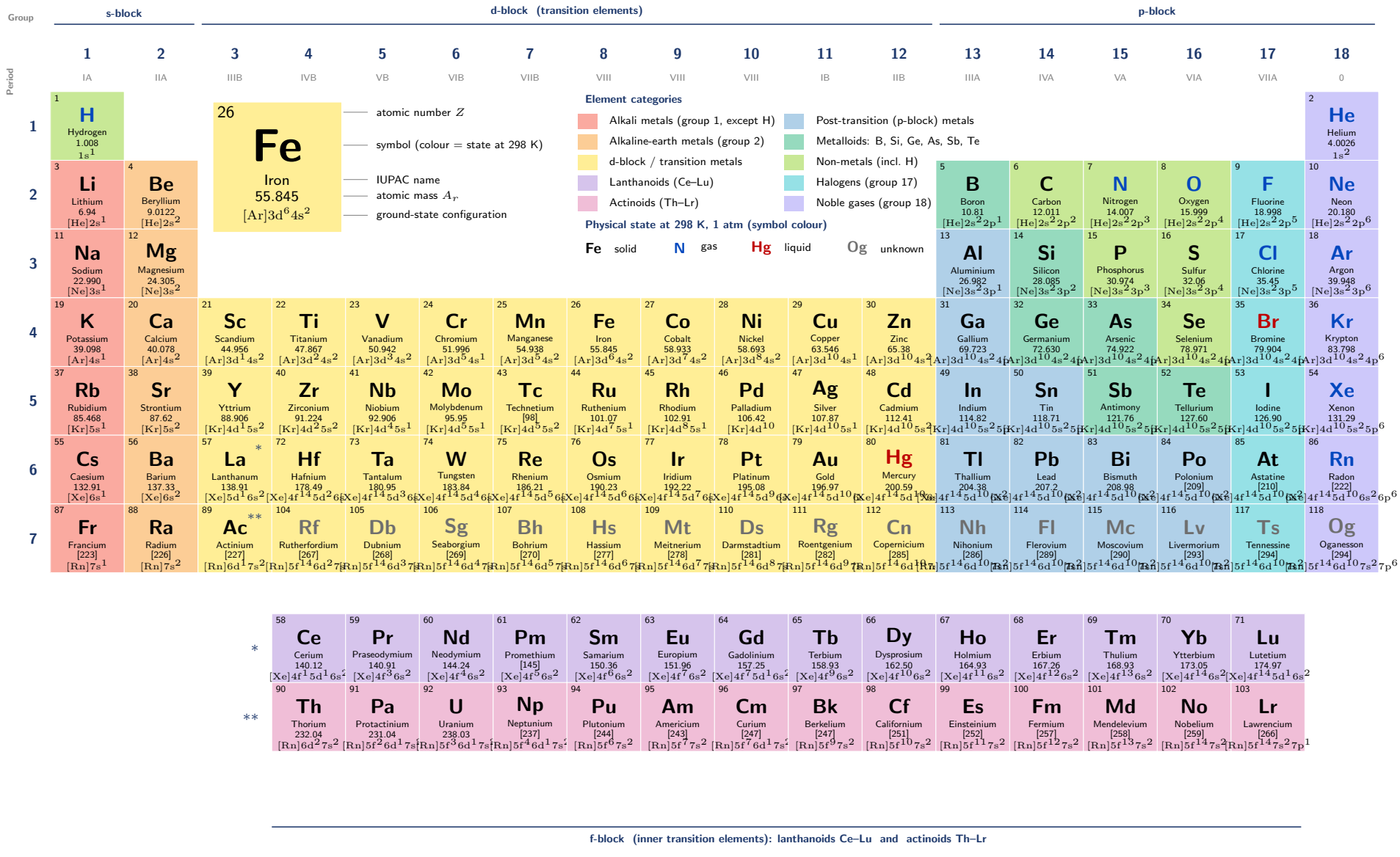


Figure 2. Schematic structure of the long-form table showing the four blocks and the general outer-electron configuration of each.

2.4 The modern periodic table of 118 elements

The following landscape page gives the complete table as recommended by IUPAC: atomic number, symbol, name, standard atomic mass, ground-state configuration, colour-coded category and physical state at 298 K.

Modern (Long-Form) Periodic Table of the 118 Elements — IUPAC Recommendation



Notes. (1) Group numbers 1–18 are the IUPAC recommendation; the older Roman-numeral labels (IA, IIA, IIIB …) are shown in grey. (2) La and Ac are shown under Sc–Y as in JEE/NCERT usage; the 2021 IUPAC provisional recommendation places Lu and Lr in group 3 (so that La–Yb and Ac–No form the f-block). (3) H is placed in group 1 because of its $1s^1$ configuration but is a non-metal; He is an s-block element placed in group 18. (4) Atomic masses are abridged standard values; a value in square brackets is the mass number of the longest-lived (or most stable) isotope. (5) Ground-state configurations are experimental for $Z \leq 103$; those for $Z \geq 104$ are predicted. Elements 113–118 were named in 2016: Nh, Mc, Ts, Og (and Fl, Lv earlier).

3 Nomenclature of Elements (IUPAC Systematic Names)

Until a newly synthesised element is confirmed and officially named, IUPAC assigns it a temporary *systematic name* derived directly from its atomic number. Each digit of Z is replaced by a numerical root and the roots are joined in order, followed by the suffix **-ium**.

Digit	0	1	2	3	4	5	6	7	8	9
Root	nil	un	bi	tri	quad	pent	hex	sept	oct	enn
Symbol letter	n	u	b	t	q	p	h	s	o	e

Formula 3.1 Systematic Name and Symbol

$$\text{Name}(Z) = \text{root}(d_1) + \text{root}(d_2) + \text{root}(d_3) + \text{-ium} \qquad \text{Symbol} = \text{initial letters of the roots}$$

Conditions and spelling rules

(i) The final *n* of *enn* is dropped when it comes before *nil* (e.g. 190 = *un-enn-nil-ium* → Unennilium). (ii) The final *i* of *bi* and *tri* is dropped before *-ium* (e.g. 102 = Unnilbium, 103 = Unniltrium). (iii) The symbol has **three** letters, the first capital. (iv) The suffix *-ium* is used for all systematic names irrespective of whether the element is expected to be a metal. (v) After IUPAC approval, the official name ends in *-ium* for metals, *-ine* for halogens (Tennessine) and *-on* for noble gases (Oganesson).

Worked Example 3.1 Systematic names of undiscovered elements

Solution. $Z = 119$: 1 → un, 1 → un, 9 → enn ⇒ **Ununennium, Uue** (group 1, period 8). $Z = 120$: un bi nil ⇒ **Unbinilium, Ubn** (group 2). $Z = 126$: un bi hex ⇒ **Unbihexium, Ubh**. $Z = 137$: un tri sept ⇒ **Untriseptium, Uts**.

Z	Systematic name	Symbol	Official name	Symbol
101	Unnilunium	Unu	Mendelevium	Md
102	Unnilbium	Unb	Nobelium	No
103	Unniltrium	Unt	Lawrencium	Lr
104	Unnilquadium	Unq	Rutherfordium	Rf
105	Unnilpentium	Unp	Dubnium	Db
106	Unnilhexium	Unh	Seaborgium	Sg
107	Unnilseptium	Uns	Bohrium	Bh
108	Unniloctium	Uno	Hassium	Hs
109	Unnilennium	Une	Meitnerium	Mt
110	Ununnilium	Uun	Darmstadtium	Ds
111	Unununium	Uuu	Roentgenium	Rg
112	Ununbium	Uub	Copernicium	Cn
113	Ununtrium	Uut	Nihonium	Nh
114	Ununquadium	Uuq	Flerovium	Fl
115	Ununpentium	Uup	Moscovium	Mc
116	Ununhexium	Uuh	Livermorium	Lv
117	Ununseptium	Uus	Tennessine	Ts
118	Ununoctium	Uuo	Oganesson	Og

Part II — Classification of Elements

4 Classification by Blocks

The elements are divided into four blocks according to the sub-shell into which the *last (differentiating) electron* enters.

Block	Groups	General outer configuration	Elements	Nature
s	1, 2 (+ H, He)	ns^{1-2}	14	Reactive metals (H and He excepted)
p	13–18	ns^2np^{1-6}	36	Metals, metalloids, non-metals
d	3–12	$(n-1)d^{1-10}ns^{0-2}$	40	All metals
f	(group 3)	$(n-2)f^{1-14}(n-1)d^{0-1}ns^2$	28	All metals; radioactive actinoids

Check: $14+36+40+28=118$. (Counts follow the table of this book: He is counted in the s-block; La and Ac in the d-block; Ce–Lu and Th–Lr form the f-block.)

4.1 s-Block elements

- The last electron enters an s orbital; an s orbital holds at most two electrons, so the general configurations are ns^1 (group 1) and ns^2 (group 2), $n=1$ to 7.
- Group 1 elements are **alkali metals** because they react with water to form alkalis (hydroxides). Group 2 are **alkaline-earth metals**: their oxides are alkaline in water and they occur in the earth's crust.
- Together they contain **12 elements** (excluding H and He). Fr and Ra are radioactive.
- Properties: soft, low melting points, very low ionisation enthalpies, strong reducing agents, characteristic flame colours (Li crimson, Na golden yellow, K lilac, Rb red-violet, Cs blue; Ca brick red, Sr crimson, Ba apple green), compounds mostly ionic.

Common Trap

The ALLEN notes call Cs and Fr "liquid". At 298 K both are *solids*, but they melt just above room temperature (Cs 301.6 K $\approx 28.5\,^{\circ}$C; Fr estimated about 300 K). The same is true of Ga (302.9 K). The only elements that are liquid at 298 K and 1 atm are **Hg** (d-block) and **Br** (p-block).

4.2 p-Block elements

- The last electron enters a p orbital. Six electrons can be accommodated, hence six groups (13, 14, 15, 16, 17, 18). General configuration ns^2np^{1-6}, $n=2$ to 7.
- Group 18 (He excepted) has ns^2np^6; the filled sub-shells make these elements inert.
- The block has **36 elements** (6 in each of periods 2–7). Older notes quote 30 or 31 (elements known up to Rn); *use the count that the question's table implies.*
- Eleven elements are gases at 298 K: H_2, He (s-block) and N_2, O_2, F_2, Cl_2, Ne, Ar, Kr, Xe, Rn (p-block, nine of them). Br is a liquid; the rest are solids.
- The staircase line in the right-hand part of the table separates metals (below left), metalloids (on the line) and non-metals (above right).

4.3 d-Block elements

- The last electron enters the $(n-1)d$ orbitals (penultimate shell), $n=4$ to 7. They lie between the s- and p-blocks, in groups 3–12 (old IIIB to IIB), forming four series: 3d (Sc–Zn), 4d (Y–Cd), 5d (La, Hf–Hg) and 6d (Ac, Rf–Cn).
- General configuration $(n-1)d^{1-10}ns^{0-2}$. All d-block elements are metals; **Hg is the only liquid**

one.

- Typical properties: variable oxidation states, coloured ions, paramagnetism, catalytic activity, complex formation, high melting points and interstitial compounds.

Anomalous configurations from extra stability of half-filled and fully filled sub-shells

Because the $(n-1)d$ and ns energies are very close, an electron moves from ns to $(n-1)d$ when this gives a half-filled (d^5) or fully filled (d^{10}) set. First series:

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
$3d$	1	2	3	**5**	5	6	7	8	**10**	10
$4s$	2	2	2	**1**	2	2	2	2	**1**	2

Other common cases: Nb $4d^4 5s^1$, Mo $4d^5 5s^1$, Ru $4d^7 5s^1$, Rh $4d^8 5s^1$, Pd $4d^{10} 5s^0$, Ag $4d^{10} 5s^1$, Pt $5d^9 6s^1$, Au $5d^{10} 6s^1$.

4.4 f-Block elements

- The last electron enters the $(n-2)f$ orbitals ($n = 6, 7$). General configuration $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$.
- They are the **lanthanoids** (Ce, $Z = 58$ to Lu, 71; they follow lanthanum, 57) and the **actinoids** (Th, 90 to Lr, 103; they follow actinium, 89): in all **28** elements.
- Lanthanoids occur in low abundance in nature and are called *rare-earth elements*. **All actinoids are radioactive**; all elements after uranium ($Z > 92$) are **transuranium** elements.
- They are also called **inner transition elements** and belong to group 3. In the atom the last shell has two electrons, the penultimate shell has 8 or 9 electrons and the pre-penultimate shell has more than 18 (up to 32) electrons.

Correct general formula

Some printed notes write the f-block formula as $(n-2)s^2p^6d^{10}f^{1-14}\ldots$, which includes complete inner shells. The usable valence form is $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$.

5 Types of Elements

5.1 Classification based on electronic configuration

Using configuration as criterion we recognise four types: the **inert (noble) gases**, the **representative (main-group) elements**, the **transition elements** and the **inner transition elements**. The class depends on the extent to which the s, p, d and f orbitals are filled.

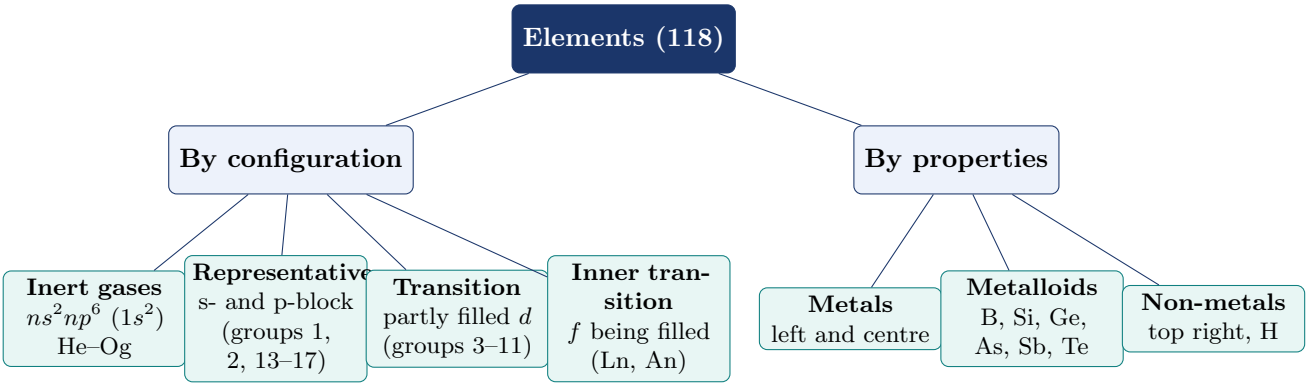


Figure 3. Two complementary ways of classifying the elements.

5.2 Inert (noble) gases

- The s and p orbitals of the outermost shell are completely filled: configuration ns^2np^6. Helium is also an inert gas but has configuration $1s^2$.
- Examples: He, Ne, Ar, Kr, Xe, Rn (and Og, predicted to be a solid at 298 K).
- Very high ionisation enthalpy, zero (positive) electron gain enthalpy, monatomic, colourless and odourless, the largest atomic (van der Waals) radius in each period. Xe and Kr do form compounds (XeF_2, XeF_4, XeO_3, KrF_2), so "inert" is now replaced by "noble".

5.3 Representative elements

All elements of the s-block (except He) and the p-block (groups 13–17). Their properties change regularly across periods, and their valency follows the group-number rules of Section 9.

5.4 Transition elements

Definition

An element is a **transition element** if it has a *partly filled d sub-shell either in the neutral atom or in any of its common oxidation states.*

Hence Zn, Cd and Hg (group 12) are d-block elements but *not* transition elements: they have d^{10} in the neutral state and in the stable +2 state. By contrast Cu is a transition element since Cu^{2+} is $3d^9$, and Sc (neutral $3d^1$) is also counted.

5.5 Inner transition elements

Lanthanoids and actinoids; the last *three* shells (outermost, penultimate and pre-penultimate) are incomplete.

5.6 Metals

The majority of elements are metals; they lie on the left and in the centre of the table.

- Usually solid at room temperature (exception: Hg).
- High melting and boiling points (exceptions: Ga and Cs, 303 K and 302 K; Na, K, Hg are also low).
- Good conductors of heat and electricity; lustrous; malleable (hammered into sheets) and ductile (drawn into wires); sonorous.
- Lose electrons to form cations; low ionisation enthalpies, low electronegativities (generally $\chi < 2$); strong reducing agents.
- Form *basic or amphoteric oxides* and ionic chlorides. Examples: Na, Mg, Fe, Cu, Zn, Al, Ag, Au.

5.7 Non-metals

Placed at the top right of the table. Along a period, properties change from metallic (left) to non-metallic (right).

- Usually solids or gases (Br is a liquid) at room temperature.
- Low melting and boiling points (exceptions: boron, carbon as diamond and graphite).
- Most non-metallic solids are brittle and neither malleable nor ductile; poor conductors of heat and electricity (exceptions: graphite conducts; iodine is lustrous).
- High ionisation enthalpy and electronegativity; gain electrons to form anions; *acidic or neutral oxides*; good oxidising agents. Examples: H, C, N, O, P, S, F, Cl, Br, I, noble gases.

5.8 Metalloids (semi-metals)

Elements with properties intermediate between metals and non-metals lie along the staircase line: **Si, Ge, As, Sb, Te** (the ALLEN list; NCERT also includes **B**, and Po and At are sometimes added).

- Metallic lustre but brittle; *semiconductors*, i.e. electrical conductivity between that of metals and insulators and increasing with temperature.
- Oxides are amphoteric (As_2O_3, Sb_2O_3) or weakly acidic (B_2O_3, SiO_2); they form covalent compounds.

Property	Metals	Non-metals	Metalloids
Ionisation enthalpy	Low	High	Intermediate
Electronegativity	Low ($\lesssim 1.8$)	High	Intermediate ($\sim$ 1.8–2.1)
Ion formed	Cation	Anion	Either (rarely)
Oxide	Basic / amphoteric	Acidic / neutral	Amphoteric / weakly acidic
Conduction	Good conductors	Insulators (graphite excepted)	Semiconductors
Redox behaviour	Reducing agents	Oxidising agents	Either
Examples	Na, Al, Fe, Cu	C, N, O, Cl	Si, Ge, As, Sb, Te

Part III — Locating an Element in the Table

6 Predicting the Atomic Number of Successive Members of a Group

6.1 Magic numbers

Knowing the atomic number of the first member of a group, the atomic numbers of the following members are obtained by adding the appropriate *magic number* (the number of elements in a period: 2, 8, 8, 18, 18, 32, 32).

Derivation. Let an element of group g lie in period p. The next element below it in the same group has the same outer configuration one shell higher, so it lies exactly one period further down. Counting elements along the path:

- **Groups 1 and 2 (s-block):** from the element, the remaining $N_p -$ (position) elements of period p are crossed, and then the first (position) elements of period $p+1$ are reached; the net number is the *length of the period of the upper element*: $\Delta Z = N_p$. Example: H (period 1) $\to$ Li: $\Delta Z = N_1 = 2$; Li (period 2) $\to$ Na: $N_2 = 8$.
- **Groups 13–18 (p-block) and groups 3–12 (d-block):** the new element sits after the d-/f-inserts of the next period, so $\Delta Z = N_{p+1}$, the *length of the period of the lower element*. Example: Cl (17) $\to$ Br (35): $N_4 = 18$.

Formula 6.1 Atomic-Number Rule

$$\boxed{Z_{\text{below}} = Z_{\text{above}} + N} \qquad \begin{cases} N = N_p & \text{groups 1, 2 (period of upper element)} \\ N = N_{p+1} & \text{groups 13–18 (period of lower element)} \end{cases}$$

Conditions

$N_p = 2, 8, 8, 18, 18, 32, 32$ for $p = 1, \ldots, 7$. For the d-block, $\Delta Z = 18$ from period 4 to 5, $\Delta Z = 32$ from period 5 to 6 and from 6 to 7; group 3 is special (Sc 21, Y 39, La 57 (+18 each), Ac 89 (+32)). H and He are first members of groups 1 and 18 (H $\to$ Li: +2; He $\to$ Ne: +8).

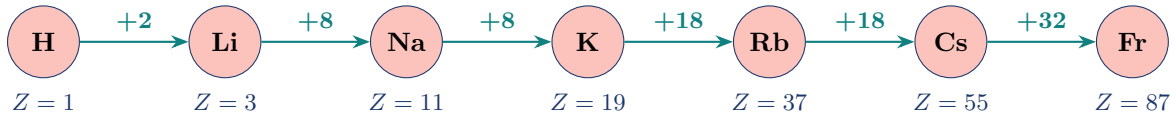


Figure 4. Group 1: atomic numbers from the magic numbers $2, 8, 8, 18, 18, 32$ ($1+2=3$, $3+8=11$, $11+8=19$, $19+18=37$, $37+18=55$, $55+32=87$).

Group(s)	Sequence of increments (down the group)	Example
1	2, 8, 8, 18, 18, 32	H 1 → Li 3 → Na 11 → K 19 → Rb 37 → Cs 55 → Fr 87
2	8, 8, 18, 18, 32	Be 4 → Mg 12 → Ca 20 → Sr 38 → Ba 56 → Ra 88
3	18, 18, 32	Sc 21 → Y 39 → La 57 → Ac 89
4–12	18, 32, 32	Ti 22 → Zr 40 → Hf 72 → Rf 104
13–17	8, 18, 18, 32, 32	F 9 → Cl 17 → Br 35 → I 53 → At 85 → Ts 117
18	8, 8, 18, 18, 32, 32	He 2 → Ne 10 → Ar 18 → Kr 36 → Xe 54 → Rn 86 → Og 118

(The d-block entries of groups 4–12 start from period 4; in the p-block of groups 13–17 the first member is in period 2.)

Group	P1	P2	P3	P4	P5	P6	P7
1	1 H	3 Li	11 Na	19 K	37 Rb	55 Cs	87 Fr
2	–	4 Be	12 Mg	20 Ca	38 Sr	56 Ba	88 Ra
3	–	–	–	21 Sc	39 Y	57 La	89 Ac
4	–	–	–	22 Ti	40 Zr	72 Hf	104 Rf
5	–	–	–	23 V	41 Nb	73 Ta	105 Db
6	–	–	–	24 Cr	42 Mo	74 W	106 Sg
7	–	–	–	25 Mn	43 Tc	75 Re	107 Bh
8	–	–	–	26 Fe	44 Ru	76 Os	108 Hs
9	–	–	–	27 Co	45 Rh	77 Ir	109 Mt
10	–	–	–	28 Ni	46 Pd	78 Pt	110 Ds
11	–	–	–	29 Cu	47 Ag	79 Au	111 Rg
12	–	–	–	30 Zn	48 Cd	80 Hg	112 Cn
13	–	5 B	13 Al	31 Ga	49 In	81 Tl	113 Nh
14	–	6 C	14 Si	32 Ge	50 Sn	82 Pb	114 Fl
15	–	7 N	15 P	33 As	51 Sb	83 Bi	115 Mc
16	–	8 O	16 S	34 Se	52 Te	84 Po	116 Lv
17	–	9 F	17 Cl	35 Br	53 I	85 At	117 Ts
18	2 He	10 Ne	18 Ar	36 Kr	54 Xe	86 Rn	118 Og

Table 1. Atomic numbers and symbols by group and period (group 3 shows La and Ac; the f-block is omitted).

Worked Example 6.1 Which group does $Z = 114$ belong to?

Solution. Carbon family: C 6, Si $6+8=14$, Ge $14+18=32$, Sn $32+18=50$, Pb $50+32=82$, next $82+32=114$. So element 114 (Flerovium) is in **group 14** and resembles Pb. Likewise $Z=119$ ($87+32$) would be a group-1 element and $Z=120$ a group-2 element (period 8).

7 Determination of Period, Block and Group of an Element

(a) Period number = principal quantum number n of the valence (outermost) shell, i.e. the highest n in the configuration.

(b) Block = the type of sub-shell (s, p, d, f) that receives the last electron.

(c) Group number is obtained from the electrons in the valence shell and the penultimate shell, as follows.

Formula 7.1 Group Number from the Configuration

s-block: $\boxed{\text{group} = n_s}$

p-block: $\boxed{\text{group} = 10 + n_s + n_p} = 12 + n_p$

d-block: $\boxed{\text{group} = n_{(n-1)d} + n_{ns}}$

f-block: $\boxed{\text{group 3 (III B)}}$

Conditions

n_s, n_p are the numbers of electrons in ns, np of the *valence shell*; use the actual (anomalous) configuration, e.g. Cr $3d^5 4s^1$ gives $5 + 1 = 6$. Valid for the d-block up to $d^{10}s^2$ (group 12). For He ($1s^2$) the formula gives group 2 but He is placed in group 18. The classical rule that the p-block group is $10 + n_s + n_p$ counts the filled $(n-1)d^{10}$ implicitly.

S.No.	Electronic configuration	Period	Block	Group by rule	Element
1	[Ar] $3d^{10}4s^2 4p^6$, $5s^1$	5	s	1	Rb
2	[Kr] $5s^2 4d^{10} 5p^2$	5	p	$2 + 10 + 2 = 14$	Sn
3	[Rn] $7s^2 5f^{14} 6d^4$	7	d	$2 + 4 = 6$	Sg
4	[Xe] $6s^2 5d^1 4f^{12}$	6	f	3 (IIIB)	lanthanoid
5	[Ar] $3d^5 4s^1$	4	d	$5 + 1 = 6$	Cr

7.1 Step-by-step procedure from the atomic number

Step 1. Locate the period using the noble-gas atomic numbers $2, 10, 18, 36, 54, 86, 118$: the period is the first one whose last element has $Z \geq Z_{\text{given}}$.

Step 2. Write the configuration (core noble gas + Madelung order), correcting for known anomalies (d^5, d^{10}).

Step 3. Name the block from the last sub-shell and compute the group with the formula above.

Worked Example 7.1 Position of $Z = 35$, 44, 74, 58 and 118

Z	Configuration	Period	Block	Group	Element
35	[Ar]$3d^{10}4s^2 4p^5$	4	p	$10 + 2 + 5 = 17$	Br (halogen)
44	[Kr]$4d^7 5s^1$	5	d	$7 + 1 = 8$	Ru
74	[Xe]$4f^{14}5d^4 6s^2$	6	d	$4 + 2 = 6$	W
58	[Xe]$4f^1 5d^1 6s^2$	6	f	3	Ce (lanthanoid)
118	[Rn]$5f^{14}6d^{10}7s^2 7p^6$	7	p	$10 + 2 + 6 = 18$	Og (noble gas)

7.2 Illustrations

Worked Example 7.2 Can an element with $Z = 126$ be placed in the present table?

Solution. No. Element 126 would begin to fill the $5g$ sub-shell ($7s^2 8s^2 \ldots 5g$ order), and the present long form has no provision for a g-block, which would require an eighth period.

Worked Example 7.3 Why does the third period contain only 8 elements, not 18?

Solution. The number of elements in a period equals twice the number of orbitals filled in it. In the third period only $3s$ and $3p$ (four orbitals) are filled before the energy of $4s$ falls below that of

$3d$, so $4 \times 2 = 8$ elements.

Worked Example 7.4 Which well-known element would $Z = 114$ resemble?

Solution. Its configuration is $[\mathrm{Rn}]5f^{14}6d^{10}7s^27p^2$, i.e. ns^2np^2: group 14, the carbon family (resembles Pb).

Part IV — Shielding and Periodicity

8 Screening Effect and Effective Nuclear Charge

8.1 Concept

A valence electron is attracted by the nucleus (charge $+Ze$) and repelled by the electrons lying between it and the nucleus (inner-shell electrons) and, to a smaller extent, by the other valence electrons. The decrease in the attraction on the valence electron caused by inner-shell electrons is called the **screening (shielding) effect**. Because of it the valence electron experiences a *reduced* nuclear charge, called the **effective nuclear charge**, Z_{eff} (or Z^*).

Derivation. The Coulomb force on a valence electron at distance r from the nucleus is

$$F = \frac{(Z\,e)(e)}{4\pi\varepsilon_0 r^2} - \frac{\sigma\,e\cdot e}{4\pi\varepsilon_0 r^2} = \frac{(Z-\sigma)\,e^2}{4\pi\varepsilon_0\,r^2}$$

where each unit of screening σ cancels one unit of the nuclear charge. The attraction therefore behaves as though the nuclear charge were $Z - \sigma$.

Formula 8.1 Effective Nuclear Charge

$$\boxed{Z_{\text{eff}} = Z - \sigma}$$

Conditions

Z is the nuclear charge (atomic number); σ (sigma) is the **screening constant**, the total screening contributed by all other electrons, calculated for *one particular electron* (normally the outermost one). Z_{eff} is therefore different for different electrons of the same atom.

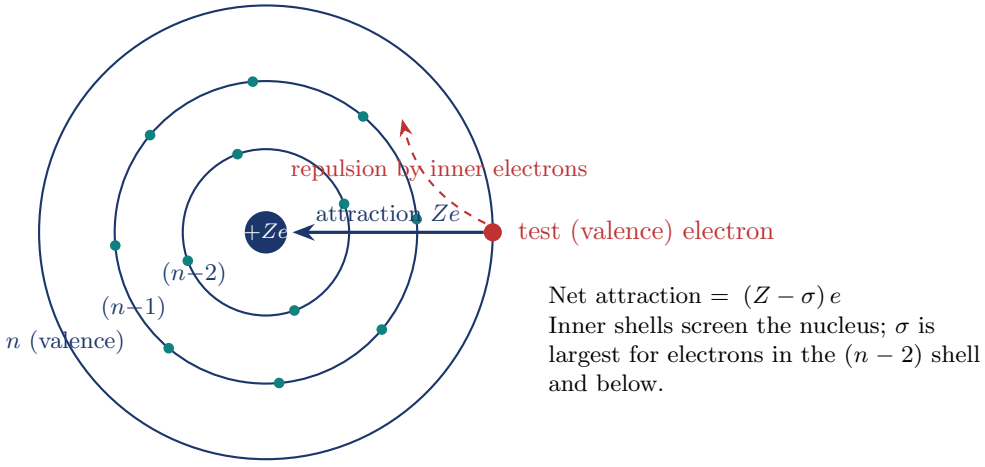


Figure 5. Screening of the nuclear charge by inner-shell electrons.

8.2 Calculation of σ: Slater's rules

Step 1. Write the electronic configuration in the following order and groupings:

$$(1s)\,(2s,2p)\,(3s,3p)\,(3d)\,(4s,4p)\,(4d)\,(4f)\,(5s,5p)\,(5d)\,(5f)\ldots$$

Step 2. Electrons in groups to the *right* of the group containing the electron of interest contribute **nothing**.

Step 3. Apply the values below to the remaining electrons.

Electron of interest	Same group	$(n-1)$ shell	$(n-2)$ shell and lower
ns or np	0.35 each (0.30 for $1s$)	0.85 each	1.00 each
nd or nf	0.35 each	1.00 each (all groups to the left)	1.00 each

Formula 8.2 Slater's Screening Constant

$$\boxed{\sigma = 0.35\,N_{\text{same}} + 0.85\,N_{(n-1)} + 1.00\,N_{\leq(n-2)}} \quad (ns,\ np) \qquad \boxed{\sigma = 0.35\,N_{\text{same}} + 1.00\,N_{\text{left}}} \quad (nd,\ nf)$$

Conditions

N_{same} counts the *other* electrons in the same group (for a $1s$ electron the other $1s$ electron counts 0.30). Slater's rules give approximate values; they are exact enough to explain periodic trends, but not exact energies. The d electrons of the $(n-1)$ shell screen an ns electron by 0.85 each, i.e. *poorly*, and this is the origin of the d-block contraction.

Worked Example 8.1 Z_{eff} across the second period (ALLEN table)

For Li to F: the two $1s$ electrons are in the $(n-1)$ shell, contributing $2 \times 0.85 = 1.70$ to every element; the other $2s, 2p$ electrons contribute 0.35 each.

Element	Config.	Z	σ of ns, np	σ of $(n-1)$	Total σ	$Z_{\text{eff}} = Z - \sigma$
Li	$1s^2 2s^1$	3	–	1.70	1.70	1.30
Be	$1s^2 2s^2$	4	$1 \times 0.35 = 0.35$	1.70	2.05	1.95
B	$1s^2 2s^2 2p^1$	5	$2 \times 0.35 = 0.70$	1.70	2.40	2.60
C	$1s^2 2s^2 2p^2$	6	$3 \times 0.35 = 1.05$	1.70	2.75	3.25
N	$1s^2 2s^2 2p^3$	7	$4 \times 0.35 = 1.40$	1.70	3.10	3.90
O	$1s^2 2s^2 2p^4$	8	$5 \times 0.35 = 1.75$	1.70	3.45	4.55
F	$1s^2 2s^2 2p^5$	9	$6 \times 0.35 = 2.10$	1.70	3.80	5.20
Ne	$1s^2 2s^2 2p^6$	10	$7 \times 0.35 = 2.45$	1.70	4.15	5.85

Worked Example 8.2 Z_{eff} in the third period

For Na to Ar the $(n-1)$ shell is $2s^2 2p^6$ (8 electrons $\times 0.85 = 6.80$) and the $(n-2)$ shell is $1s^2$ ($2 \times 1.00 = 2.00$): fixed contribution 8.80.

Element	Na	Mg	Al	Si	P	S	Cl	Ar
σ	8.80	9.15	9.50	9.85	10.20	10.55	10.90	11.25
Z_{eff}	2.20	2.85	3.50	4.15	4.80	5.45	6.10	6.75

Worked Example 8.3 Z_{eff} for N^-, N and N^+ (ALLEN Illustration)

Solution. The last electron is a $2p$ electron.

$$\begin{aligned} \text{N } (1s^2 2s^2 2p^3) &: \sigma = 4(0.35) + 2(0.85) = 3.10, & Z_{\text{eff}} &= 7 - 3.10 = 3.90 \\ \text{N}^- \ (1s^2 2s^2 2p^4) &: \sigma = 5(0.35) + 2(0.85) = 3.45, & Z_{\text{eff}} &= 7 - 3.45 = 3.55 \\ \text{N}^+ \ (1s^2 2s^2 2p^2) &: \sigma = 3(0.35) + 2(0.85) = 2.75, & Z_{\text{eff}} &= 7 - 2.75 = 4.25 \end{aligned}$$

Hence Z_{eff}: $\text{N}^- < \text{N} < \text{N}^+$, which is why anions are larger and cations smaller than the parent atom.

Worked Example 8.4 Z_{eff} of the $4s$ and $3d$ electrons of iron

Fe: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$. For a **4s** electron: same group 1×0.35; $(n-1) = 3$ shell has $2+6+6 = 14$ electrons $\times 0.85 = 11.90$; $(n-2)$ and lower $= 10 \times 1.00$. $\sigma = 22.25$, $Z_{\text{eff}} = 26 - 22.25 = \mathbf{3.75}$. For a **3d** electron: same group $5 \times 0.35 = 1.75$; all electrons to the left ($1s, 2s, 2p, 3s, 3p$: 18) $\times 1.00$; the $4s$ electrons to the right give 0. $\sigma = 19.75$, $Z_{\text{eff}} = 26 - 19.75 = \mathbf{6.25}$. **Solution.** The $4s$ electrons feel a much smaller Z_{eff} than $3d$ electrons, so they are lost first on ionisation, though $4s$ is filled first.

8.3 Periodic variation of Z_{eff}

Formula 8.3 Change of Z_{eff} with Atomic Number

$\boxed{\Delta Z_{\text{eff}} = 1 - 0.35 = 0.65}$ (across a period, s, p) $\boxed{\Delta Z_{\text{eff}} = 1 - 0.85 = 0.15}$ (across a d-series, ns electron)

Reasoning and conditions

(i) Across a period (representative elements) Z rises by 1 while the new electron enters the same shell and screens the others by only 0.35, so Z_{eff} rises by 0.65 per element and the atomic size falls considerably. (ii) In a transition series the new electron enters the $(n-1)d$ sub-shell and screens the outer ns electron by 0.85, so Z_{eff} rises by only $1 - 0.85 = 0.15$; the size therefore remains almost constant. (iii) Down a main group, Z_{eff} for the valence electron remains *almost constant* (Li 1.30 is the exception, since its inner shell is $1s$): inner shells are added, each screening fully.

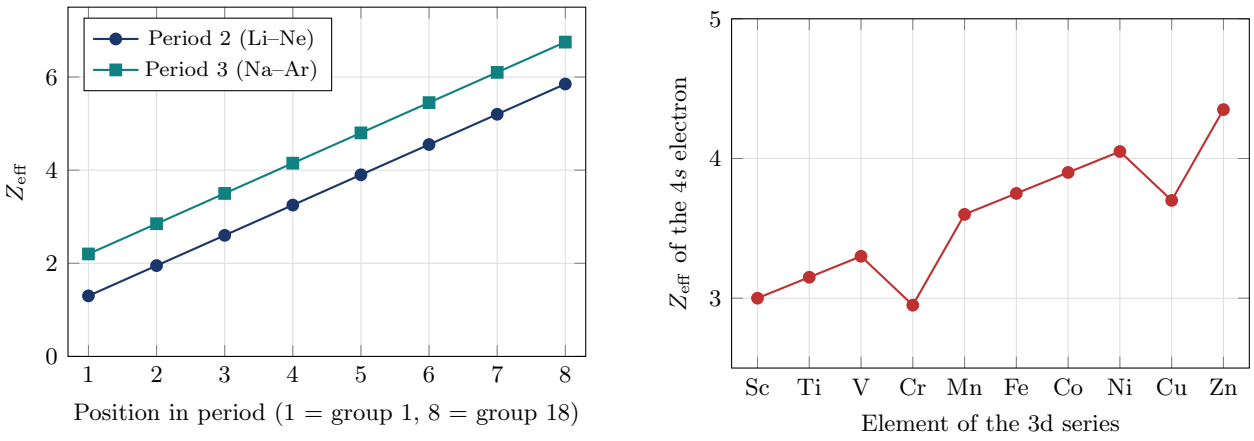


Figure 6. Left: Z_{eff} rises by 0.65 per element across periods 2 and 3. Right: Z_{eff} of the $4s$ electron across 3d rises by only 0.15; dips occur at Cr ($3d^5 4s^1$) and Cu ($3d^{10} 4s^1$) because $4s^1$ has no same-group screening partner.

Group 13: a hint of the d-block contraction

Slater's rule for the np^1 electron gives Z_{eff}: B 2.60, Al 3.50, **Ga 5.00**, In 5.00, Tl 5.00. The jump from Al to Ga arises because the ten $3d$ electrons screen only 0.85 each while the nuclear charge rises by 18; as a result Ga is hardly larger than Al.

Chromium

The ALLEN transition-series list gives Cr = 3.45, which assumes the hypothetical $3d^4 4s^2$ configuration. For the real ground state $3d^5 4s^1$ the Slater value is $\sigma = 0 + 13 \times 0.85 + 10 = 21.05$, $Z_{\text{eff}} = 24 - 21.05 = 2.95$.

9 Periodicity

9.1 Meaning and causes

The regular gradation of properties from top to bottom in a group and the repetition of similar properties after definite intervals of atomic number is called **periodicity**. In a period the number of electrons in the outermost shell rises from 1 to 8 while the shell number is the same; in a group the number of valence electrons is the same while n increases.

Cause of periodicity

The periodicity of properties arises from the **recurrence of similar outer-shell electronic configurations at regular intervals**. Elements with similar properties occur after atomic-number intervals of $2, 8, 8, 18, 18, 32$, the *magic numbers*.

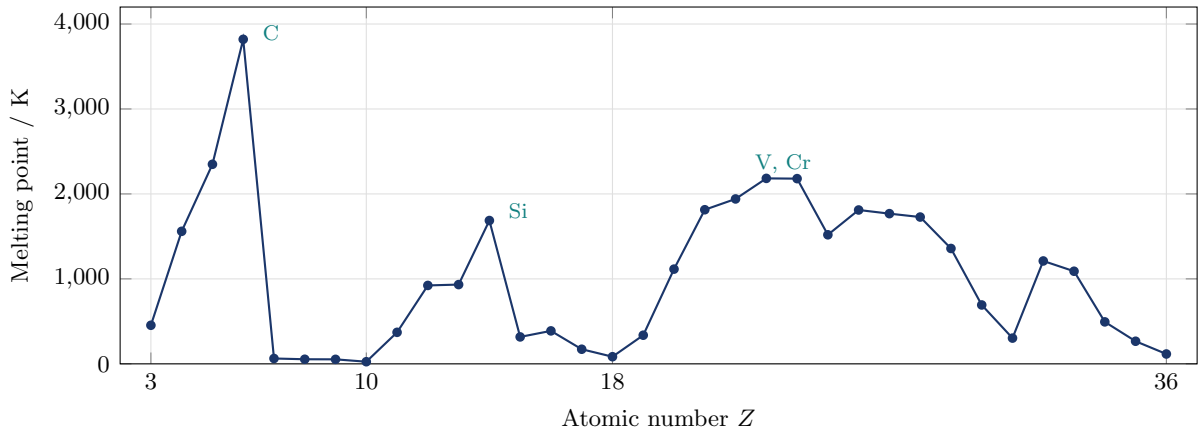


Figure 7. Periodicity of a physical property: melting points of the elements Li to Kr ($Z = 3$–36). Maxima fall in the centre of each period, minima at the noble gases.

9.2 Periodic properties

Type	Properties
Atomic / structural	Atomic radius, ionic radius, atomic volume, density, Z_{eff}
Energetic	Ionisation enthalpy, electron gain enthalpy, electronegativity, lattice and hydration enthalpies
Physical	Melting and boiling points, metallic character, conductivity, hardness
Chemical	Valency, oxidation states, reducing/oxidising power, acidic–basic nature of oxides and hydroxides, nature of hydrides

9.3 Valency

Valency is the combining capacity of an element (Italian *valentia*, capacity). Frankland introduced the old concept.

9.3.1 Valency with respect to hydrogen (old concept)

Valency of H = 1. It is the number of hydrogen atoms that combine with one atom of the element.

Group	1	2	13	14	15	16	17
Hydride	NaH	MgH_2	AlH_3	SiH_4	PH_3	H_2S	HCl
Valency	1	2	3	4	3	2	1

Valency w.r.t. H increases from 1 to 4 across a period and then decreases to 1.

9.3.2 Valency with respect to oxygen

Valency of O = 2. It is *twice the number of oxygen atoms combining with one atom of the element.*

Group	1	2	13	14	15	16	17
Oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_2O_5	SO_3	Cl_2O_7
Valency	1	2	3	4	5	6	7

Valency with respect to oxygen increases from 1 to 7 across the period, and equals the (old) group number.

9.3.3 Electronic concept

Formula 9.1 Valency from the Valence-Shell Electrons

$$\boxed{\text{Valency} = n_{\text{val}}} \quad (\text{groups } 1, 2, 13, 14) \qquad \boxed{\text{Valency} = 8 - n_{\text{val}}} \quad (\text{groups } 15, 16, 17, 18)$$

Conditions

n_{val} is the number of electrons in the valence shell. The rule gives the *covalent* (hydride-type) valency; maximum valencies with oxygen follow the group number. All elements of a group have the same valency because they have the same number of valence electrons.

Group	1	2	13	14	15	16	17	18
Configuration	ns^1	ns^2	ns^2np^1	ns^2np^2	ns^2np^3	ns^2np^4	ns^2np^5	ns^2np^6
Valence electrons	1	2	3	4	5	6	7	8
Valency	1	2	3	4	3	2	1	0

Part V — Periodic Properties

10 Atomic Radius

10.1 Definition and difficulty

The atomic radius is the average distance of the valence-shell electrons from the nucleus. It cannot be measured directly because (i) a single isolated atom is very hard to study and (ii) an atom has no sharp boundary (the probability of finding the electron falls to zero only at infinity). The operational definition is therefore based on *half of an internuclear distance* (bond length d), and the value depends on the kind of bond between the atoms.

10.2 How to calculate the atomic radius

10.2.1 Covalent radius from bond length

Derivation. In a homonuclear diatomic molecule A–A joined by a single bond the two nuclei are a distance $d_{\mathrm{A-A}}$ apart, and the two identical atoms share this distance equally.

Formula 10.1 Covalent Radius

$$\boxed{r_{\mathrm{A}} = \frac{d_{\mathrm{A-A}}}{2}} \qquad \boxed{d_{\mathrm{A-B}} = r_{\mathrm{A}} + r_{\mathrm{B}}} \quad (\chi_A \approx \chi_B)$$

Conditions

The bond must be a *single* covalent bond (radii shrink with multiple bonding: C–C 77, C=C 67, C≡C 60 pm). Additivity is exact only for non-polar bonds.

For a heteronuclear molecule AB with different electronegativities the bond is shortened by its ionic character; the experimental length is *less* than $r_A + r_B$ (Schomaker–Stevenson correction):

Formula 10.2 Schomaker–Stevenson Equation

$$\boxed{d_{\mathrm{A-B}} = r_{\mathrm{A}} + r_{\mathrm{B}} - 0.09\,|\chi_{\mathrm{A}} - \chi_{\mathrm{B}}|} \quad (d, r \text{ in Å})$$

Worked Example 10.1 H–Cl bond length

$r_{\mathrm{H}} = 0.37$ Å and $r_{\mathrm{Cl}} = 0.99$ Å; $\chi_{\mathrm{H}} = 2.1$, $\chi_{\mathrm{Cl}} = 3.0$. **Solution.** $d_{\mathrm{H-Cl}} = 0.37 + 0.99 - 0.09(3.0 - 2.1) = 1.36 - 0.081 = \mathbf{1.28}$ Å (experimental 1.27 Å).

Worked Example 10.2 ALLEN illustration

$d_{\mathrm{Cl-Cl}} = 10$ Å, $d_{\mathrm{H-H}} = 2$ Å (given). Then $r_{\mathrm{Cl}} = 5$ Å, $r_{\mathrm{H}} = 1$ Å; $d_{\mathrm{H-Cl}} = 1 + 5 - 0.09(3.0 - 2.1) = \mathbf{5.919}$ Å.

Worked Example 10.3 C–X bond length (ALLEN)

X–X is 1.00 Å, C–C is 1.54 Å; $\chi_{\mathrm{X}} = 3.0$, $\chi_{\mathrm{C}} = 2.0$. **Solution.** $d_{\mathrm{C-X}} = \frac{1.00}{2} + \frac{1.54}{2} - 0.09(1.0) = 0.50 + 0.77 - 0.09 = \mathbf{1.18}$ Å.

10.2.2 Metallic radius from the crystal lattice

Metal atoms are treated as hard spheres in contact; half the distance between nearest neighbours is the **metallic radius**.

Derivation. In a face-centred cubic (fcc) cell of edge a the atoms touch along the face diagonal, whose length is $a\sqrt{2} = 4r$. In a body-centred cubic (bcc) cell the atoms touch along the body diagonal $a\sqrt{3} = 4r$. In a simple cubic cell they touch along the edge: $a = 2r$.

Formula 10.3 Metallic Radius from Edge Length a

$$\boxed{r_{\mathrm{fcc}} = \frac{a}{2\sqrt{2}}} \qquad \boxed{r_{\mathrm{bcc}} = \frac{\sqrt{3}\,a}{4}} \qquad \boxed{r_{\mathrm{sc}} = \frac{a}{2}}$$

Worked Example 10.4 Copper (fcc, $a = 361$ pm) and sodium (bcc, $a = 429$ pm)

Solution. $r_{\mathrm{Cu}} = \dfrac{361}{2\sqrt{2}} = \mathbf{127.6}$ pm; $\quad r_{\mathrm{Na}} = \dfrac{\sqrt{3}(429)}{4} = \mathbf{185.8}$ pm.

10.2.3 van der Waals radius

In the solid state non-metals exist as molecules held together by van der Waals forces. **One half of the distance between the nuclei of two adjacent atoms belonging to two neighbouring molecules** (non-bonded atoms) is the van der Waals (collision) radius.

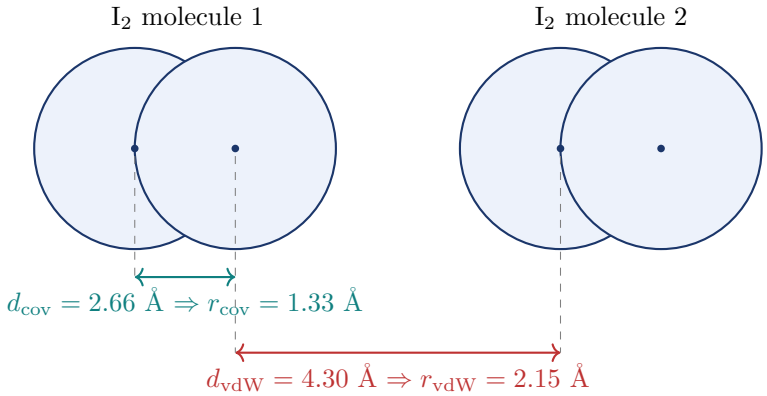


Figure 8. Covalent and van der Waals radii of iodine: $r_{\mathrm{vdW}} > r_{\mathrm{cov}}$.

10.2.4 Ionic radius: Pauling's method

Derivation. In an isoelectronic pair of ions, the electron cloud of each ion is held by $Z_{\mathrm{eff}} = Z - \sigma$; the larger the Z_{eff}, the smaller the ion. Pauling therefore took $r \propto 1/Z_{\mathrm{eff}}$ and divided the measured cation–anion distance d in the ratio of the Z_{eff} values:

$$\frac{r_+}{r_-} = \frac{Z_- - \sigma}{Z_+ - \sigma}, \qquad r_+ + r_- = d$$

Formula 10.4 Pauling Ionic Radii

$$\boxed{r_+ = \frac{d\,(Z_- - \sigma)}{(Z_+ - \sigma) + (Z_- - \sigma)}} \qquad \boxed{r_- = d - r_+} \qquad (\sigma = 4.52 \text{ for the Ne core})$$

Conditions

Applies to a pair of isoelectronic ions in contact in an ionic crystal (same σ for both). Radii depend on the coordination number (they increase with CN) and on the radius-scale used.

Worked Example 10.5 Na^+ and F^- in NaF ($d = 231$ pm)

Solution. $\dfrac{r_+}{r_-} = \dfrac{9 - 4.52}{11 - 4.52} = \dfrac{4.48}{6.48} = 0.691$. Hence $r_{\mathrm{F^-}} = \dfrac{231}{1.691} = \mathbf{136.6}$ pm and $r_{\mathrm{Na^+}} = 231 - 136.6 = \mathbf{94.4}$ pm (tabulated 136 and 95 pm).

10.3 Types of atomic radius

Radius	Definition	Elements	Example
Covalent	Half the internuclear distance of two atoms joined by a *single* covalent bond in a homonuclear molecule	Non-metals, metalloids	Cl: 99 pm, I: 133 pm
Metallic	Half the distance between nearest atoms in a metallic crystal	Metals	K: 231 pm, Na: 186 pm
van der Waals	Half the distance between nuclei of non-bonded atoms of neighbouring molecules in the solid	Non-metals, noble gases	Cl: 180 pm, I: 215 pm
Ionic	Radius of the ion in an ionic crystal (from $r_+ + r_- = d$)	Ions	Na^+: 95 pm, F^-: 136 pm

Formula 10.5 Order of Radii for the Same Atom

$$r_{\text{vdW}} > r_{\text{metallic}} > r_{\text{covalent}}$$

Reason

Metal atoms in a metallic lattice overlap less strongly than atoms in a covalent bond *(K: metallic 231 pm vs covalent 203 pm)*; van der Waals contacts involve no bonding overlap at all (Cl: 180 pm vs 99 pm). Hence van der Waals radius is the largest, and noble gases (for which only this radius is defined) have the largest "radius" in a period.

10.4 Ionic radii

10.4.1 Cation

The radius of a cation is *smaller* than that of the parent atom: (i) either a whole shell is removed (Na $\rightarrow$ Na^+), or (ii) the proton-to-electron ratio increases, so Z_{eff} and attraction increase, and interelectronic repulsion decreases.

10.4.2 Anion

The radius of an anion is *larger* than that of the parent atom: Z_{eff} decreases (the number of protons is unchanged but electrons increase) and interelectronic repulsion increases.

10.4.3 Isoelectronic species

Formula 10.6 Size of Isoelectronic Species

$$r \propto \frac{1}{Z_{\text{eff}}} \propto \frac{1}{Z} \quad \text{(same number of electrons)}$$

Ne-type (10 e^-)	N^{3-}	O^{2-}	F^-	Na^+	Mg^{2+}	Al^{3+}
Z	7	8	9	11	12	13
Z_{eff} (Slater, 2p)	2.85	3.85	4.85	6.85	7.85	8.85
Radius / pm (Pauling)	171	140	136	95	65	50
Ar-type (18 e^-)	**P^{3-}**	**S^{2-}**	**Cl^-**	**K^+**	**Ca^{2+}**	**Sc^{3+}**
Radius / pm (Pauling)	212	184	181	133	99	81

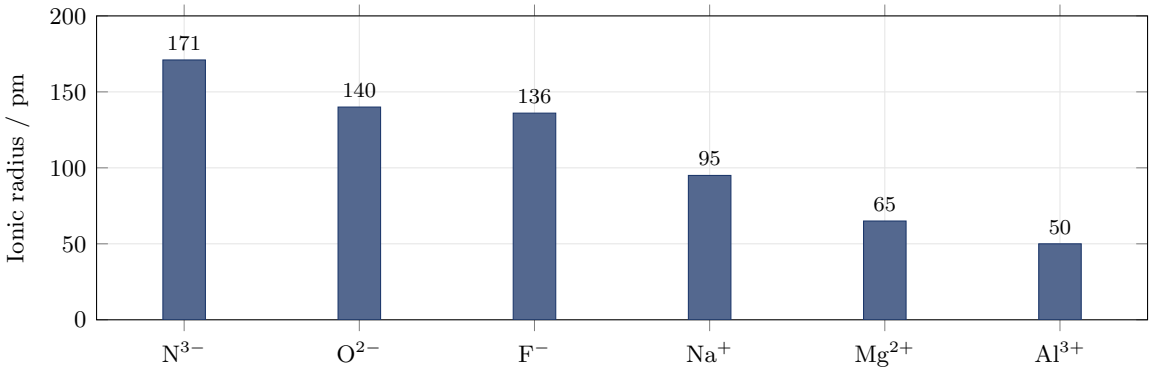


Figure 9. Radii of the isoelectronic Ne-type ions decrease as the nuclear charge increases.

10.5 Factors affecting atomic radius

Factor	Effect on radius	Illustration
(a) Z_{eff} increases	Radius decreases	Li > Be > B > C > N > O > F
(b) Number of shells n increases	Radius increases	Li < Na < K < Rb < Cs
(c) Screening effect increases	Radius increases	Inner shells added down a group
(d) Magnitude of negative charge increases	Radius increases	$O < O^- < O^{2-}$
(e) Magnitude of positive charge increases	Radius decreases	$Mn > Mn^{2+} > Mn^{3+} > Mn^{4+}$
(f) Bond order / coordination number	Radius decreases for multiple bonds; increases with CN	C–C 77, C=C 67, C≡C 60 pm

10.6 Periodic trend

10.6.1 Representative elements

Across a period the atomic radius decreases from left to right because Z_{eff} increases (Section 8): Li > Be > B > C > N > O > F. **Down a group** it increases because the number of shells increases: Li < Na < K < Rb < Cs.

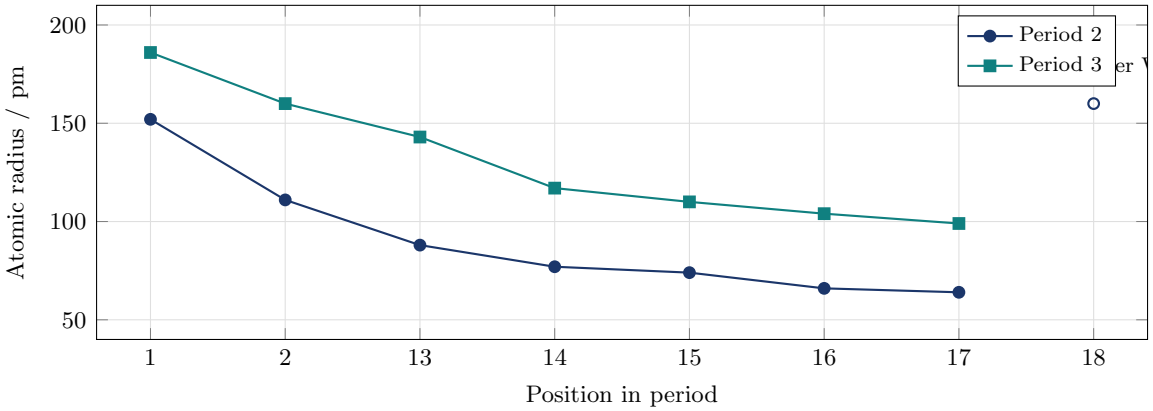


Figure 10. Radius decreases across periods 2 and 3 (metallic radius for metals, covalent for non-metals); the noble gases have van der Waals radii and appear larger.

Group 13: Al and Ga

Radius of Al (143 pm) is slightly *larger* than that of Ga (135 pm). From Al to Ga, 18 electrons are added, 10 of them in 3*d* orbitals which shield the nuclear charge poorly; the outer electrons of Ga are pulled in more strongly. Such contraction owing to poor shielding by *d* electrons is called the

transition (or d-block) contraction.

10.6.2 Noble gases

Only the van der Waals radius is defined, so in each period the noble gas is the *largest* atom; the radius also increases down the group.

10.6.3 Transition elements

- **Across a series:** in the 3d series the radius decreases from Sc to Mn because the increasing Z_{eff} dominates; it then remains almost constant (Fe, Co, Ni, Cu) because screening by added d electrons balances the extra nuclear charge ($\Delta Z_{\text{eff}} = 0.15$); from Cu to Zn the size increases slightly because of the d^{10} configuration (electron–electron repulsion and shielding dominate).
- **Down a group:** 3d → 4d: radius increases (extra shell). 4d → 5d: radii are *almost the same* (Zr 160, Hf 159; Nb 146, Ta 146; Mo 139, W 139 pm), the exceptions being group 3 (Y 180, La 187 pm), because of the lanthanide contraction.

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Radius/pm	162	147	134	127	126	126	124	124	124	138

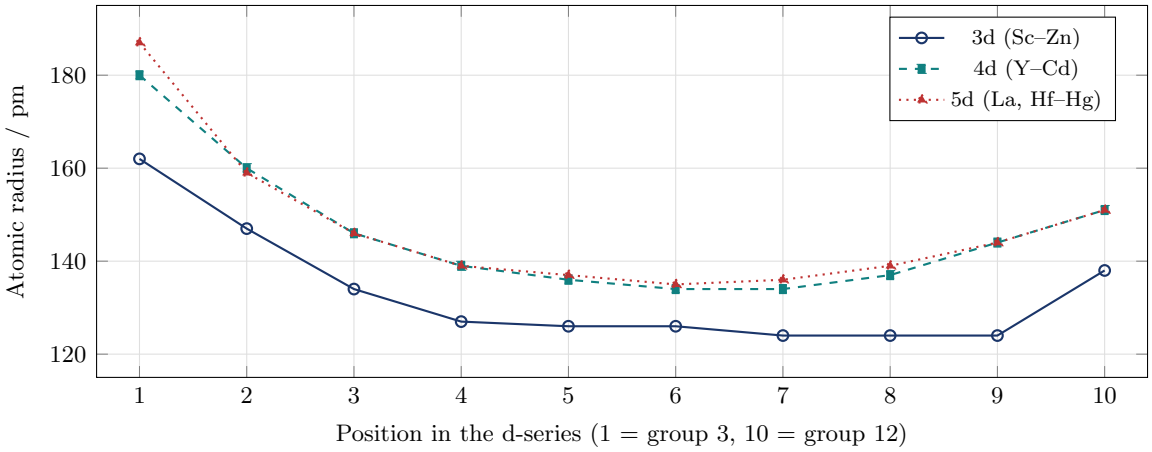


Figure 11. Trends in atomic radii of the transition series (representative values). The 4d and 5d curves almost coincide from Zr/Hf onward (lanthanide contraction).

10.6.4 Inner transition elements: the lanthanoid contraction

Along the lanthanoid series (La to Lu) there is a steady decrease in atomic and ionic radius; the decrease is regular for ions but less regular for atoms.

Derivation. Across the series the nuclear charge rises by +1 at each step while the added electron enters the inner $(n-2)$ $4f$ sub-shell. The $4f$ electrons have diffuse, "poorly shielding" shapes, so the $6s$ electrons (which fix the size) feel a slowly increasing Z_{eff}. The decrease per element is small, but over 14 elements it accumulates to about 13 pm in metallic radius (La 187 pm → Lu 174 pm).

Formula 10.7 Lanthanoid Contraction

$$\boxed{Z_{\text{eff}}(\text{Lu}) > Z_{\text{eff}}(\text{La}) \;\Rightarrow\; r_{\text{Lu}} < r_{\text{La}}} \quad (\approx 13 \text{ pm})$$

Consequences of lanthanoid contraction

(i) Hf (5d) has almost the same radius as Zr (4d), so Zr/Hf, Nb/Ta, Mo/W are chemically alike and very difficult to separate. (ii) The increase of +32 in nuclear charge from the 4d to the 5d element balances the extra shell; 5d metals therefore have high densities and high ionisation enthalpies. (iii) Basic strength of $Ln(OH)_3$ decreases from La to Lu. (iv) Atoms of Hf to Hg following Lu are smaller than expected. (v) Electronegativity of group 12 rises down the group.

Other values worth remembering

Free H^- ion: 208 pm; in LiH and NaH: 130 and 137 pm (radii vary with coordination number). Ionic size *increases* in the direction of the arrows in the diagonal pairs Li^+–Na^+–K^+ and F^-–Cl^-–Br^-–I^-.

Worked Example 10.6 Order of radii of V^{5+}, K^+, S^{2-}, Mn^{7+}, Ca^{2+}, Cl^-, P^{3-}

Solution. All have 18 electrons ($Z = 23, 19, 16, 25, 20, 17, 15$) and are isoelectronic with argon. For isoelectronic species $r \propto 1/Z$:

$$Mn^{7+} < V^{5+} < Ca^{2+} < K^+ < Cl^- < S^{2-} < P^{3-}$$

Worked Example 10.7 Smallest ion among Li^+, Be^{2+}, B^{3+} and the radii of Cu, Cu^+, Cu^{2+}

Solution. All of Li^+, Be^{2+}, B^{3+} have the $1s^2$ configuration; B^{3+} has the highest Z (hence Z_{eff}) and is the **smallest**. For copper, successive loss of electrons increases Z_{eff} per electron, so the radius order Cu (122 pm) > Cu^+ (96 pm) > Cu^{2+} (72 pm) reported by the student is correct.

11 Ionisation Enthalpy

Ionisation energy is the minimum energy required to remove the most loosely bound electron from an *isolated gaseous atom* in its ground state. Expressed in electron-volts it is the *ionisation potential*; per mole of atoms it is the *ionisation enthalpy*, $\Delta_i H$.

Formula 11.1 First Ionisation Enthalpy

$$\boxed{X(g) \longrightarrow X^+(g) + e^- \qquad \Delta_i H = IE_1 > 0}$$

Conditions

The atom must be isolated (gaseous, no bonding to other atoms) and in its ground state; a solid or molecule must first be converted to gaseous atoms (sublimation or atomisation energy is *not* part of IE). Ionisation is *always endothermic* ($\Delta H = +$ve). Units: kJ mol^{-1}, kcal mol^{-1} or eV/atom, with 1 eV = 96.485 kJ mol^{-1} = 23.06 kcal mol^{-1}.

11.1 How to calculate ionisation enthalpy

11.1.1 (a) One-electron (hydrogen-like) species: Bohr model

Derivation. The energy of the electron in orbit n of a species of nuclear charge Z is $E_n = -\dfrac{2.18 \times 10^{-18}\, Z^2}{n^2}$ J. Removing the electron to $n = \infty$ ($E_\infty = 0$) requires IE $= 0 - E_n$:

Formula 11.2 Ionisation Energy of H-like Species

$$\boxed{\mathrm{IE} = 2.18 \times 10^{-18} \frac{Z^2}{n^2}\ \mathrm{J\,atom^{-1}} = 13.6 \frac{Z^2}{n^2}\ \mathrm{eV} = 1312 \frac{Z^2}{n^2}\ \mathrm{kJ\,mol^{-1}}}$$

Conditions

Valid only for one-electron species (H, He^+, Li^{2+}, ...) because the formula ignores electron–electron repulsion. For many-electron atoms use measured values or Slater-type reasoning to compare, never to compute.

Worked Example 11.1 ALLEN illustration

The energy to remove the electron from one H atom is 2.18×10^{-18} J. Per mole: $2.18 \times 10^{-18} \times 6.02 \times 10^{23} = 1.31 \times 10^{6}\ \mathrm{J\,mol^{-1}} = \mathbf{1310\ kJ\,mol^{-1}}$. For He^+: $13.6 \times 4 = 54.4$ eV; Li^{2+}: $13.6 \times 9 = 122.4$ eV.

11.1.2 (b) From a photon (photo-ionisation threshold)

Derivation. A photon of frequency ν ionises an atom if $h\nu \geq \mathrm{IE}$; at threshold $h\nu_0 = \mathrm{IE}$ (kinetic energy of electron zero). Multiplying by N_A:

Formula 11.3 IE from Threshold Wavelength

$$\boxed{\mathrm{IE} = \frac{N_A h c}{\lambda_0}} \quad \text{i.e.} \quad \boxed{\mathrm{IE\ (kJ\,mol^{-1})} = \frac{1.196 \times 10^5}{\lambda_0\ \mathrm{(nm)}}}$$

Worked Example 11.2 Sodium

$\mathrm{IE_1(Na)} = 496\ \mathrm{kJ\,mol^{-1}}$, so $\lambda_0 = \dfrac{1.196 \times 10^5}{496} = \mathbf{241}$ nm. Light of wavelength *shorter* than 241 nm ionises gaseous Na; the excess energy appears as kinetic energy of the electron: $h\nu = \mathrm{IE} + \frac{1}{2}mv^2$.

11.1.3 (c) Unit conversion and energy for a given mass

Formula 11.4 Energy to Ionise a Sample

$$\boxed{E = \frac{m}{M} \sum_{i=1}^{k} \mathrm{IE}_i} \quad \text{(to form } \mathrm{M^{k+}}\text{)}$$

Worked Example 11.3 ALLEN: 12 mg of Mg vapour → Mg^{2+}

$\mathrm{IE_1} = 7.646$ eV, $\mathrm{IE_2} = 15.035$ eV, 1 eV $= 96.5\ \mathrm{kJ\,mol^{-1}}$. **Solution.** $\mathrm{IE_1} + \mathrm{IE_2} = 22.681$ eV $= 2188.6\ \mathrm{kJ\,mol^{-1}}$; $n = \dfrac{12 \times 10^{-3}}{24} = 0.5 \times 10^{-3}$ mol; $E = 0.5 \times 10^{-3} \times 2188.6 = \mathbf{1.094\ kJ}$.

11.2 Successive ionisation enthalpies

For an atom M the successive ionisation enthalpies are defined by

$$\mathrm{M} + \mathrm{IE_1} \rightarrow \mathrm{M^+} + e^-, \qquad \mathrm{M^+} + \mathrm{IE_2} \rightarrow \mathrm{M^{2+}} + e^-, \qquad \mathrm{M^{2+}} + \mathrm{IE_3} \rightarrow \mathrm{M^{3+}} + e^-$$

Formula 11.5 Order of Successive Ionisation Enthalpies

$$\boxed{\mathrm{IE}_1 < \mathrm{IE}_2 < \mathrm{IE}_3 < \cdots}$$

Reasons

After each removal the number of electrons decreases while Z is unchanged, so Z_{eff} per electron and the attraction increase; the electron is also taken from a progressively smaller cation. **A very large jump** between IE_k and IE_{k+1} shows that the $(k+1)$th electron comes from a completed inner shell, i.e. the element has k valence electrons.

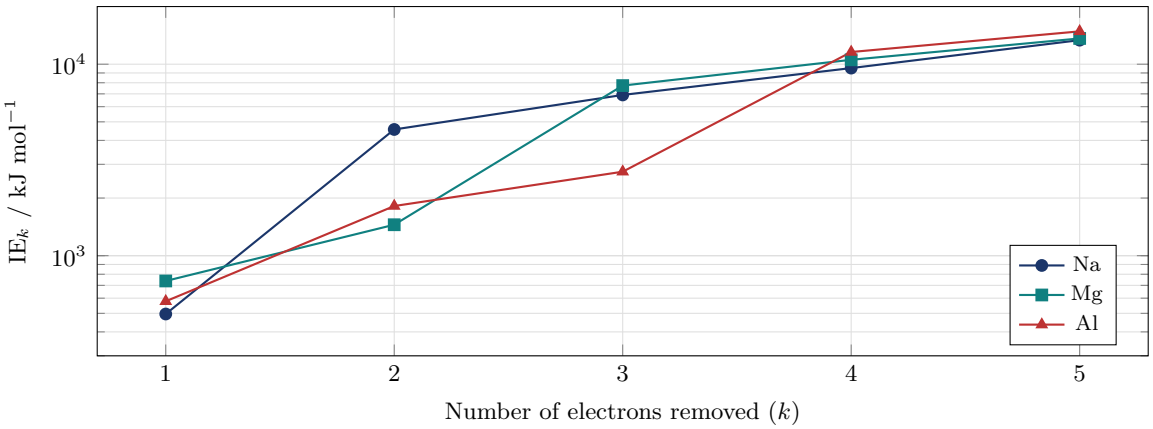


Figure 12. Successive ionisation enthalpies of Na, Mg, Al. The big jump occurs after $k = 1, 2, 3$ respectively, which is the number of valence electrons.

Worked Example 11.4 Beginner's Box problems: identify the element from IE data

(i) Successive IE of X (kcal mol^{-1}): 165, 195, 556, 595. The large jump is between IE_2 and IE_3: two valence electrons, configuration $1s^2 2s^2 2p^6 3s^2$ (Mg), answer (D). (ii) IE_1–IE_5 = 7.1, 14.3, 34.5, 46.8, 162.2 eV: the jump is after IE_4, hence four valence electrons (ns^2np^2): **Si**. (iii) Mg forms Mg(II) because $\mathrm{IE}_2 - \mathrm{IE}_1$ is below 11 eV while $\mathrm{IE}_3 - \mathrm{IE}_2$ is above 16 eV (see thumb rule below).

11.3 Factors affecting ionisation enthalpy

11.3.1 (a) Atomic size

The larger the atom, the farther the outermost electron is from the nucleus, the weaker the attraction and the lower the ionisation enthalpy.

Formula 11.6 Atomic Size

$$\boxed{\mathrm{IE} \propto \frac{1}{\text{atomic size}}}$$

11.3.2 (b) Effective nuclear charge

Formula 11.7 Effective Nuclear Charge

$$\boxed{\mathrm{IE} \propto Z_{\text{eff}}}$$

11.3.3 (c) Screening effect

The greater the screening by inner electrons, the lesser the attraction on the outer electron, hence the lower the ionisation enthalpy.

Formula 11.8 Screening Effect

$$\text{IE} \propto \frac{1}{\text{screening effect}}$$

11.3.4 (d) Penetration power of sub-shells

The penetration of orbitals into the core follows $s > p > d > f$ (for the same n). An s electron is held more tightly, so more energy is needed to remove it than a p electron of the same shell.

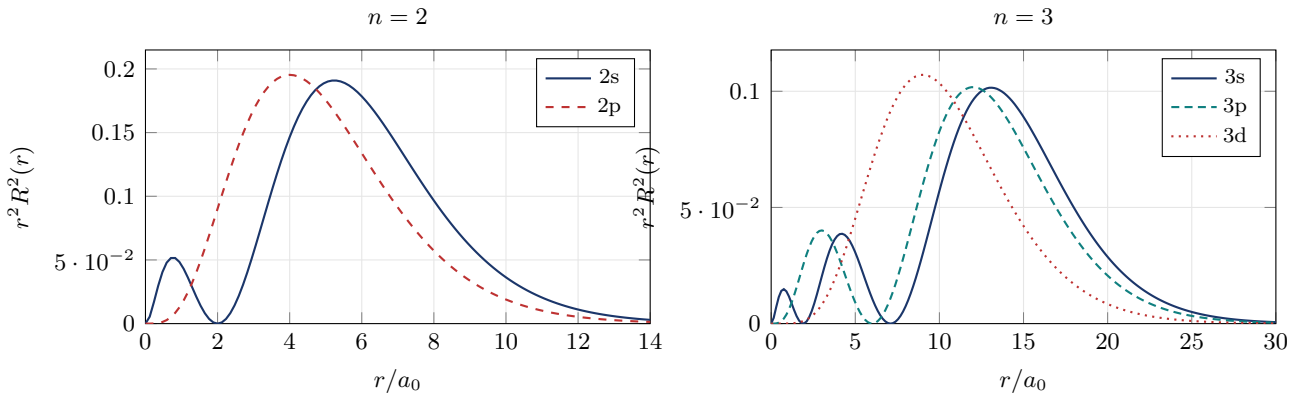


Figure 13. Radial probability (hydrogen-like orbitals): the s orbital has inner maxima, so it penetrates the core; the sharp outer maximum of p and d lies in the screened region. Energy order for the same n: $ns < np < nd < nf$.

Worked Example 11.5 Be vs B, and IE_2 of B vs Be

Be ($1s^2 2s^2$) loses a penetrating $2s$ electron; B ($1s^2 2s^2 2p^1$) loses a less penetrating $2p$ electron. Hence IE_1: Be (899) > B (801). After losing one electron, B^+ is $1s^2 2s^2$ (stable, filled s like Be), so IE_2: B (2427) > Be (1757).

11.3.5 (e) Stability of half-filled and fully filled orbitals

Half-filled (p^3, d^5, f^7) and fully filled (s^2, p^6, d^{10}, f^{14}) configurations have extra stability (exchange energy and symmetry), so extra energy is needed to remove an electron from them.

- N ($2p^3$, half-filled) has higher IE_1 than O ($2p^4$): **O** < **N**. Similarly S < P.
- Mg ($3s^2$ filled, high penetration) has higher IE_1 than Al ($3p^1$): **Na** < **Al** < **Mg**.
- IE_2 order: Mg^+ ($3s^1$, i.e. 2,8,1) < Al^+ ($3s^2$, 2,8,2) < Na^+ (2,8; noble-gas core): Na^+ is hardest to ionise.

11.4 Periodic trend

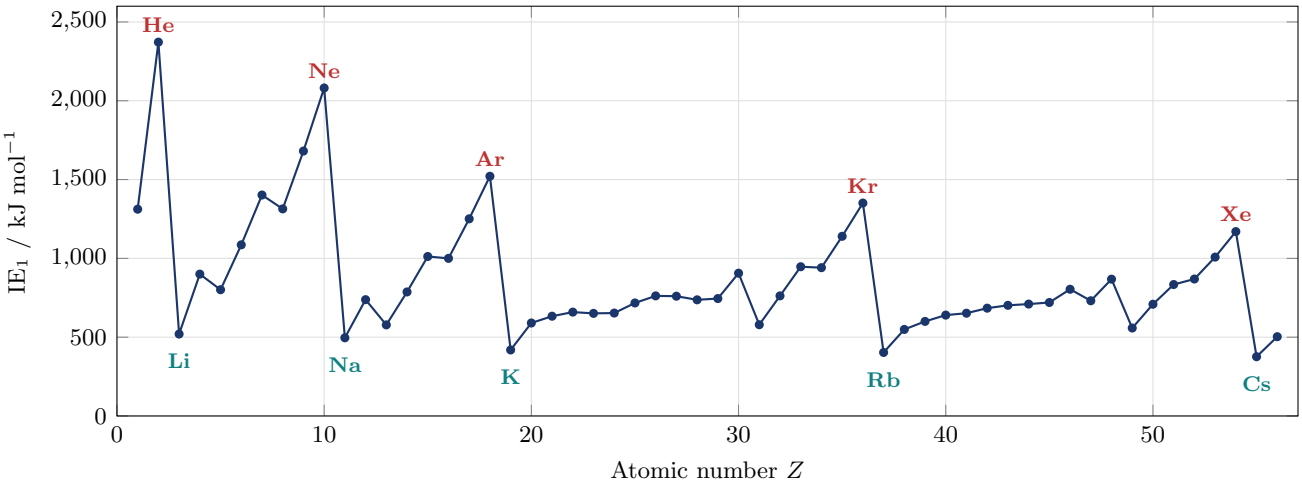


Figure 14. First ionisation enthalpy for $Z = 1$–56. Peaks: noble gases; troughs: alkali metals. He has the highest IE_1; Cs the lowest among the stable elements.

11.4.1 Variation in a period among the representative elements

IE_1 generally increases from left to right: Z_{eff} increases while the principal quantum number remains the same, so the size decreases.

Period 2	Li	Be	B	C	N	O	F	Ne
IE_1/kJ mol^{-1}	520	899	801	1086	1402	1314	1681	2081
Period 3	Na	Mg	Al	Si	P	S	Cl	Ar
IE_1/kJ mol^{-1}	496	738	578	787	1012	1000	1251	1521

Order in period 2: **Li** < **B** < **Be** < **C** < **O** < **N** < **F** < **Ne**; period 3: **Na** < **Al** < **Mg** < **Si** < **S** < **P** < **Cl** < **Ar**.

11.4.2 Variation in a group among the representative elements

IE_1 decreases on moving down because the size increases (larger n) while Z_{eff} of the outer electron remains almost constant.

11.4.3 Variation in a period among the transition elements

IE_1 increases only slightly and irregularly across a series because the added electron goes to the inner $(n-1)d$ sub-shell, Z_{eff} rising by only ≈ 0.15 per element (Sc 633, Zn 906 kJ mol^{-1}).

11.4.4 Variation in a group among the transition elements

From the first to the second transition series IE_1 decreases (size increases); from the second to the third it increases (lanthanoid contraction).

Group	3	4	6	10	11	12
3d element	Sc 633	Ti 659	Cr 653	Ni 737	Cu 745	Zn 906
4d element	Y 600	Zr 640	Mo 684	Pd 804	Ag 731	Cd 868
5d element	La 538	Hf 659	W 770	Pt 870	Au 890	Hg 1007

11.4.5 Exceptions

(a) Group 2 > group 13: Be ($1s^2 2s^2$) > B, and Mg > Al, because the s electron penetrates more than the p electron.

(b) Group 15 > group 16: N > O, P > S, As > Se, because of the half-filled p^3 configuration; in the group 16 atoms (p^4) loss of one electron leads to the stable p^3.

(c) Group 13: Al (578) and Ga (579) have nearly equal IE_1 because of the transition (d-block) contraction; In (558) is lower, but Tl (589) is *higher* than In because the $4f$ and $5d$ electrons shield poorly.

(d) Group 14: Pb (716) is slightly higher than Sn (709): the $4f$ electrons added before the sixth-period elements do not shield the outer electrons completely, so the +32 increase in nuclear charge raises IE_1.

(e) Group 3: on moving from the second to the third transition series IE rises, except for IIIB where Y (600) > La (538).

(f) Helium has the highest IE_1 of all elements and caesium the lowest among stable ones. Noble gases have very high IE (He 24.6, Ne 21.6, Ar 15.8, Kr 14.0, Xe 12.1, Rn 10.7 eV), because of the stable s^2p^6 configuration.

(g) For isoelectronic species, ionisation enthalpy increases with positive charge and decreases with negative charge: $Al^{3+} > Mg^{2+} > Na^+ > F^- > O^{2-} > N^{3-}$.

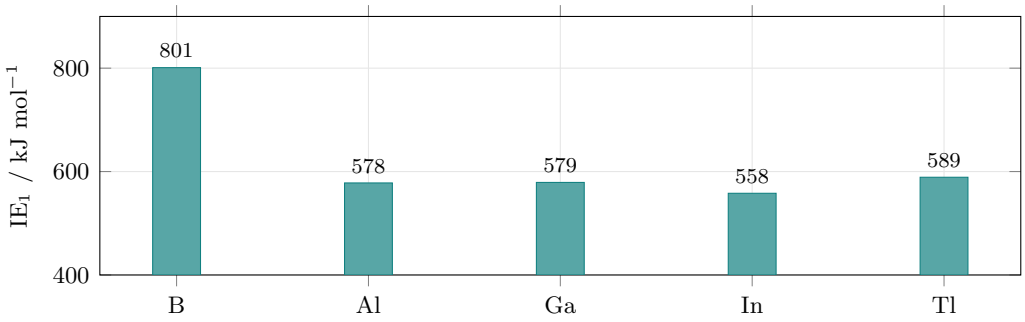


Figure 15. Group 13: the decrease down the group is interrupted at Ga and Tl by the poor shielding of d and f electrons.

11.5 Applications of ionisation enthalpy

11.5.1 Metallic and non-metallic character

Formula 11.9 Metallic Character

$$\text{IE} \propto \frac{1}{\text{metallic character}}$$

Metals (Na, K, Rb, ...) have low ionisation enthalpy, non-metals (F, Cl, Br, ...) have high ionisation enthalpy.

11.5.2 Reducing character

Formula 11.10 Reducing Power

$$\text{Reducing character} \propto \frac{1}{\text{IE}}$$

- *Gaseous state:* group 1 has the minimum IE and the strongest reducing character: Li < Na < K < Rb < Cs.
- *Aqueous state:* the reducing character follows the standard electrode potential, Li > K ≈ Rb > Cs > Na, because Li^+ has the highest hydration enthalpy (high charge density): E° (V): Li −3.05, K

-2.93, Rb ≈ -2.93, Cs ≈ -2.92, Na -2.71.

- Group 17 has the maximum ionisation potential and is a strong oxidising agent: F > Cl > Br > I (E°: 2.87, 1.36, 1.09, 0.54 V).

11.5.3 Stability of oxidation states

Formula 11.11 Thumb Rule for Oxidation State

$$\boxed{\mathrm{IE}_{k+1} - \mathrm{IE}_k \;\geq\; 16 \text{ eV} \;\Rightarrow\; \text{oxidation state } +k \text{ is stable}}$$

$$\boxed{\mathrm{IE}_{k+1} - \mathrm{IE}_k \;\leq\; 11 \text{ eV} \;\Rightarrow\; \text{state } +(k{+}1) \text{ is attained}}$$

Conditions

An empirical (JEE) criterion: a very large jump means the next electron lies in a filled core, so ionisation stops at $+k$; a small difference means the next electron is removed easily (compensated by lattice or hydration enthalpy), so the *higher* oxidation state is formed.

Element	IE_1	IE_2	IE_3	IE_4	Conclusion
Na	5.1	47.3	71.6	–	$IE_2-IE_1 = 42.2$ eV: Na^+ stable
Mg	7.6	15.0	80.1	–	$IE_2-IE_1 = 7.4$ eV $<$ 11: Mg^{2+}; $IE_3-IE_2 = 65$ eV: stops at +2
Al	6.0	18.8	28.4	120.0	$IE_4-IE_3 = 91.6$ eV: Al^{3+} (stable only in solid, liquid state)

(Values in eV.) In the gas phase Al^+ is stable; in solid and aqueous phases, hydration and lattice enthalpy make Al^{3+} the stable state.

Worked Example 11.6 Identify the elements from IE_1 and IE_2 (kJ mol^{-1})

(A) 520, 7300 (B) 1680, 3380 (C) 2370, 5250 (D) 900, 1760. **Solution.** (A) $IE_2 \gg IE_1$: an alkali metal (Li), the most reactive metal. (B) very high IE_1 but not extreme: F, the most reactive non-metal. (C) extremely high IE_1: a noble gas (He). (D) IE_2 not much greater than IE_1: an alkaline-earth metal (Be) forming a stable halide AX_2.

12 Electron Gain Enthalpy (Electron Affinity)

Electron gain enthalpy, ΔH_{eg}, is the enthalpy change when one mole of electrons is added to one mole of isolated gaseous atoms (in the ground state) to form gaseous negative ions. It measures the ease with which an atom accepts an electron and the tightness with which the electron is held. The **electron affinity** (EA) is the energy *released* in the same process.

Formula 12.1 First Electron Gain Enthalpy and Electron Affinity

$$\boxed{\mathrm{X(g)} + e^- \longrightarrow \mathrm{X^-(g)} \qquad \mathrm{EA}_1 = -\Delta H_{\mathrm{eg}_1}}$$

Sign convention

The thermodynamic convention: negative $\Delta H_{\mathrm{eg}} \Leftrightarrow$ energy released (exothermic) $\Leftrightarrow$ positive EA. Example: $\mathrm{F(g)} + e^- \rightarrow \mathrm{F^-(g)}$, $\Delta H = -328$ kJ mol^{-1} and EA $= +328$ kJ mol^{-1}; for N: $\Delta H = +31$

(older value) i.e. EA $= -31$. The two signs are opposite: read the question for the word *enthalpy* or *affinity*.

12.1 How to calculate electron gain enthalpy

12.1.1 (a) From the energy released by many atoms

Formula 12.2 Electron Gain Enthalpy from Measured Energy

$$\boxed{\Delta H_{\text{eg}} = -\frac{E_{\text{released}}}{N}\,N_A} \qquad \boxed{\Delta H_{\text{eg}}\ (\text{eV atom}^{-1}) = \frac{\Delta H_{\text{eg}}\ (\text{kJ mol}^{-1})}{96.5}}$$

Worked Example 12.1 ALLEN: iodine

Energy released when 10^6 atoms of I(g) are converted completely into I^-(g): 5.0×10^{-13} J.

Solution. $\Delta H_{\text{eg}} = -\dfrac{5.0 \times 10^{-13} \times 6.023 \times 10^{23}}{10^6} = -3.01 \times 10^5\ \text{J mol}^{-1} = \mathbf{-301\ kJ\,mol^{-1}}$, i.e. $\dfrac{-301}{96.5} = \mathbf{-3.12\ eV}$ per atom.

12.1.2 (b) From a cycle (indirect determination)

Derivation. EA is rarely measured directly; it is obtained from Hess's law applied to a process involving ionisation enthalpy. If the gaseous atoms K and F form separated K^+ and F^- ions, the enthalpy change is $\Delta H = \text{IE(K)} + \Delta H_{\text{eg}}\text{(F)}$.

Formula 12.3 EA from an Ion-Pair Cycle

$$\boxed{\Delta H = \text{IE(K)} - \text{EA(F)} \ \Rightarrow\ \text{EA(F)} = \text{IE(K)} - \Delta H}$$

Worked Example 12.2 ALLEN: $K + F \to K^+F^-$, $\Delta H = 19$ kcal mol^{-1}, IE(K) $= 4.3$ eV

Solution. $19\ \text{kcal mol}^{-1} = 19 \times 4.184\ \text{kJ mol}^{-1} = 79.5\ \text{kJ mol}^{-1} = \dfrac{79.5}{96.485} = 0.824$ eV. Hence EA(F) $= 4.3 - 0.824 = \mathbf{3.476\ eV}$ ($\Delta H_{\text{eg}} = -3.476$ eV).

12.1.3 (c) Photodetachment and Born–Haber cycle

EA can also be found from the threshold photon energy needed to detach the electron from X^- (EA $= h\nu_0$), and from the Born–Haber cycle for an ionic solid when the lattice enthalpy is known.

12.2 Successive electron gain enthalpies

The first electron gain enthalpy is usually negative. When a second electron is forced into an anion it must overcome electrostatic repulsion, so energy must be *supplied*.

Formula 12.4 Successive Electron Gain Enthalpies

$$\boxed{\text{A(g)} + e^- \to \text{A}^-\text{(g)} + \text{EA}_1} \qquad \boxed{\text{A}^-\text{(g)} + e^- + \text{EA}_2 \to \text{A}^{2-}\text{(g)}}$$

ΔH_{eg_1} (generally $-$ ve), ΔH_{eg_2} (always $+$ ve), $\Delta H_{\text{eg}_1} + \Delta H_{\text{eg}_2} > 0$ for O, S

Conditions

For an isolated gaseous atom: except ΔH_{eg_1}, all other electron gain enthalpies ($\Delta H_{eg_2}, \Delta H_{eg_3}, \dots$) are positive, and $|\Delta H_{eg_1}| < |\Delta H_{eg_2}|$. The formation of O^{2-}, S^{2-} in a crystal is possible only because the large lattice enthalpy more than compensates.

Worked Example 12.3 Oxygen (ALLEN)

$O(g) + e^- \rightarrow O^-(g)$: $\Delta H_{eg_1} = -141$ kJ mol^{-1} (released). $O^-(g) + e^- \rightarrow O^{2-}(g)$: $\Delta H_{eg_2} = +780$ kJ mol^{-1} (absorbed). Overall $O(g) + 2e^- \rightarrow O^{2-}(g)$: $\Delta H = -141 + 780 = \mathbf{+639\ kJ\ mol^{-1}}$.

12.3 Factors affecting electron gain enthalpy

Factor	Relation	Effect
(a) Atomic size	$\text{EA} \propto \frac{1}{\text{atomic size}}$	Smaller atoms attract the incoming electron more strongly
(b) Screening effect	$\text{EA} \propto \frac{1}{\text{screening}}$	More inner electrons reduce the attraction of the nucleus
(c) Effective nuclear charge	$\text{EA} \propto Z_{\text{eff}}$	Greater Z_{eff} makes ΔH_{eg} more negative
(d) Stable configuration	Filled/half-filled sub-shells give EA ≈ 0 or negative	Noble gases (ns^2np^6), Be, Mg (ns^2), N, P (np^3) have positive or small negative ΔH_{eg}

Elements He, Ne, Ar, ...	EA = zero (ΔH_{eg} = +ve)
Elements Be, Mg, Ca, ...	EA $\approx$ zero (ΔH_{eg} = +ve)
Elements N, P, As, ...	EA very low (little energy released)

12.4 Periodic variation of electron gain enthalpy

In a period: Z_{eff} increases and size decreases, so EA generally increases left to right and reaches its climax at the halogens (group 17). **In a group:** EA generally decreases down the group (larger size, same Z_{eff} per outer electron), *with the important exception that the second-period elements have lower EA than the third-period elements.*

Table 2. Electron gain enthalpy, ΔH_{eg} / kJ mol^{-1} (NCERT values).

Period	1	2	13	14	15	16	17	18
1	H −73							He +48
2	Li −60	Be +48	B −27	C −122	N 0	O −141	F −328	Ne +116
3	Na −53	Mg +39	Al −43	Si −134	P −72	S −200	Cl −349	Ar +96
4	K −48	Ca +29	Ga −30	Ge −116	As −78	Se −195	Br −325	Kr +96
5	Rb −47	Sr +5	In −30	Sn −116	Sb −103	Te −190	I −295	Xe +77
6	Cs −46	Ba +30	Tl −20	Pb −35	Bi −91	Po −174	At −270	Rn +68

Data differ between editions

The ALLEN table (an older data set) lists, for example, B −83, N +31, Be +66, Mg +67, Al −50; the NCERT values above are the ones used in most current examinations. The *pattern* is what matters: Be, Mg, N, noble gases are least negative or positive; Cl has the most negative ΔH_{eg} of all elements.

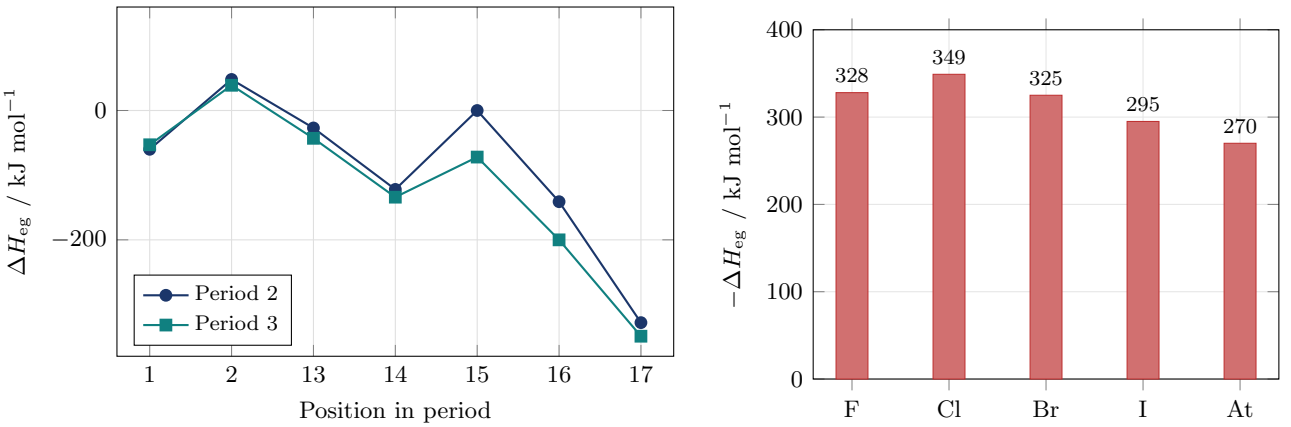


Figure 16. Left: electron gain enthalpy across periods 2 and 3 (minima at the halogens, peaks at group 2 and group 18). Right: halogens, with Cl > F.

12.4.1 Exceptions

(a) The EA of N and P (group 15) are lower than those of C and Si because of the stable half-filled np^3 configuration, and the tendency of C, Si (np^2) to attain it.

(b) The EA of zero-group elements is zero (positive ΔH_{eg}) due to the stable s^2p^6 configuration.

(c) F < Cl, O < S, N < P, B < Al in EA (the sequence is opposite to the size sequence). Though the nucleus attracts the electron more strongly in the small second-period atom, the added electron enters a very compact $2p$ sub-shell and suffers strong electron–electron repulsion; this repulsion is smaller in the larger third-period atom.

Worked Example 12.4 Why do halogens have high electron gain enthalpies?

Their configuration is ns^2np^5; one electron gives the stable noble-gas configuration ns^2np^6, so there is a strong tendency to accept an electron and a large amount of energy is released.

Worked Example 12.5 Which of O, O^-, O^{2-} has the highest electron affinity?

Solution. The neutral O atom: the incoming electron is attracted by the nucleus. For O^- and O^{2-} there is electrostatic repulsion between the added electron and the negative ion.

Worked Example 12.6 Large decrease in EA from Li to Be (ALLEN)

Li [He]$2s^1$ adds the electron to a half-vacant $2s$ orbital. Be [He]$2s^2$ must add it to the higher-energy $2p$ sub-shell, where it is poorly attracted and shielded by the filled $2s$; electron gain becomes endothermic (+48).

Worked Example 12.7 Order of negative electron gain enthalpy for N, P, O, S

Using the table: N (0) < P (−72) < O (−141) < S (−200) in increasing negative value.

13 Electronegativity

Electronegativity (χ) is a qualitative measure of the ability of an atom *in a chemical compound (bonded)* to attract the shared electrons towards itself. Unlike ionisation enthalpy and electron gain enthalpy it is not a measurable quantity, and several numerical scales (Pauling, Mulliken, Allred–Rochow) have been developed.

- In a polar bond A–B the shared pair is drawn to the more electronegative atom: $A - B \rightarrow A^- : + B^+$ if $\chi_A > \chi_B$; $A - B \rightarrow A^+ + : B^-$ if $\chi_A < \chi_B$.

- It has no unit (relative value) and is a property of a bonded atom, not of an isolated gaseous atom; no energy is liberated or absorbed.

13.1 How to calculate electronegativity

13.1.1 Pauling's scale

Derivation. Pauling observed that the bond energy E_{A-B} of a heteronuclear bond is always greater than the geometric mean of the homonuclear bond energies E_{A-A} and E_{B-B}, the excess being due to the extra stabilisation by ionic–covalent resonance. He defined this excess Δ_{AB} and found that $\sqrt{\Delta_{AB}}$ is proportional to $\chi_A - \chi_B$:

$$\Delta_{AB} = E_{A-B} - \sqrt{E_{A-A} \times E_{B-B}}$$

Formula 13.1 Pauling Electronegativity Difference

$$\boxed{\chi_A - \chi_B = 0.208\sqrt{\Delta_{AB}}} \ (\Delta \text{ in kcal mol}^{-1}) \qquad \boxed{\chi_A - \chi_B = 0.102\sqrt{\Delta_{AB}}} \ (\Delta \text{ in kJ mol}^{-1})$$

Conditions

Gives only a *difference*; the scale is fixed arbitrarily by assigning **F** = 4.0 (the element with the greatest ability to attract electrons). In the form $\chi_A - \chi_B = \sqrt{\Delta_{AB}}$ the energies are in eV ($0.208\sqrt{23.06} \approx 1.0$).

Worked Example 13.1 Electronegativity of carbon from bond energies (ALLEN)

$E_{H-H} = 104.2$, $E_{C-C} = 83.1$, $E_{C-H} = 98.8$ kcal mol^{-1}; $\chi_H = 2.1$. **Solution.** $\Delta = 98.8 - \sqrt{83.1 \times 104.2} = 98.8 - 93.05 = 5.75$; $\chi_C - 2.1 = 0.208\sqrt{5.75} = 0.208 \times 2.398 = 0.499$; $\chi_C = \mathbf{2.6}$ (tabulated 2.5).

Worked Example 13.2 Electronegativity of chlorine from ClF (ALLEN)

$E_{Cl-F} = 61$, $E_{F-F} = 38$, $E_{Cl-Cl} = 58$ kcal mol^{-1}; $\chi_F = 4$. **Solution.** $\Delta_{F-Cl} = 61 - \sqrt{38 \times 58} = 61 - 46.95 = 14.05$; $\chi_F - \chi_{Cl} = 0.208\sqrt{14.05} = 0.78$; $\chi_{Cl} = 4 - 0.78 = \mathbf{3.2}$.

13.1.2 Mulliken's scale

Derivation. An atom with high ionisation enthalpy and high electron affinity holds its own electrons strongly and attracts others strongly; Mulliken therefore defined electronegativity as the average of IE and EA of the valence state:

Formula 13.2 Mulliken Electronegativity

$$\boxed{\chi_M = \frac{IE + EA}{2}} \ (\text{eV}) \qquad \boxed{\chi_P = \frac{\chi_M}{2.8} = \frac{IE + EA}{5.6}}$$

Conditions

IE and EA must both be expressed in electron-volts; Mulliken values are about 2.8 times Pauling values. (An alternative fit $\chi_P = 0.336(\chi_M - 0.615)$ is used in some texts; the $\div 2.8$ form is the JEE standard.)

Worked Example 13.3 ALLEN: chlorine

IE of Cl^- = 4 eV (i.e. EA of Cl), EA of Cl^+ = 13.0 eV (i.e. IE of Cl). **Solution.** $\chi_P = \dfrac{4 + 13.0}{5.6} = \mathbf{3.04}$ (the printed "3.03" is a rounding difference).

13.1.3 Allred–Rochow scale

Derivation. Allred and Rochow measured electronegativity by the electrostatic force exerted by the nucleus (through Z_{eff}) on a valence electron at a distance r equal to the covalent radius, $F = \dfrac{Z_{\text{eff}} e^2}{4\pi\varepsilon_0 r^2}$. Fitting F to Pauling values gives

Formula 13.3 Allred–Rochow Electronegativity

$$\chi_{\text{AR}} = \frac{0.359\, Z_{\text{eff}}}{r^2} + 0.744 \qquad (Z_{\text{eff}} = Z - \sigma, \quad r \text{ in Å})$$

Worked Example 13.4 Li, C and Cl

Using Slater Z_{eff} and covalent radii: Li: $\dfrac{0.359 \times 1.30}{(1.34)^2} + 0.744 = 1.00$; C: $\dfrac{0.359 \times 3.25}{(0.77)^2} + 0.744 = 2.71$; Cl: $\dfrac{0.359 \times 6.10}{(0.99)^2} + 0.744 = 2.98$ — the trend Li < C < Cl agrees with Pauling (1.0, 2.5, 3.0).

13.2 Typical values (Pauling scale)

	1	2	13	14	15	16	17
1	2.1 H						
2	1.0 Li	1.5 Be	2.0 B	2.5 C	3.0 N	3.5 O	4.0 F
3	0.9 Na	1.2 Mg	1.5 Al	1.8 Si	2.1 P	2.5 S	3.0 Cl
4	0.8 K	1.0 Ca	1.6 Ga	1.8 Ge	2.0 As	2.4 Se	2.8 Br
5	0.8 Rb	1.0 Sr	1.7 In	1.8 Sn	1.9 Sb	2.1 Te	2.5 I
6	0.7 Cs	0.9 Ba	1.8 Tl	1.8 Pb	1.9 Bi	2.0 Po	2.2 At
7	0.7 Fr	0.9 Ra					

Group 18 omitted; d-block omitted (transition-metal values lie between 1.2 and 2.5). Darker colour = higher electronegativity: F (4.0) is the most electronegative element, Cs and Fr (0.7) the least.

Figure 17. Electronegativity (Pauling) of the s- and p-block elements.

13.3 Difference between electronegativity and electron affinity

S.No.	Electronegativity	Electron affinity
1	Tendency of an atom *in a molecule* to attract the bonded electrons	Energy released when an electron is added to a neutral *isolated gaseous* atom
2	Relative value	Absolute value
3	Changes regularly in a period or group	Does not change regularly
4	Has no unit	Measured in eV/atom, kJ mol^{-1} or kcal mol^{-1}

Remember

Small atoms are normally more electronegative than larger atoms. The effect of the nuclear charge balances the effect of the extra shell: **electronegativity: F > Cl, but electron affinity: Cl > F.**

13.4 Factors affecting electronegativity

(a) Charge on the atom: a cation is more electronegative than its parent atom, and χ rises with positive charge: $Fe^{3+} > Fe^{2+}$. An anion is less electronegative, and χ falls with negative charge: $O^{2-} < O^{-} < O$.

(b) Effect of substituents: the carbon atom in CF_3I carries a greater positive charge than in CH_3I, so the C of CF_3I is more electronegative.

(c) Ionisation enthalpy and electron affinity: the higher they are, the greater the electronegativity.

(d) Hybridisation: s electrons are more strongly attracted (penetrate more) than p electrons; electronegativity increases with s-character: C in CH_4 (sp^3, 25% s) < C in C_2H_4 (sp^2, 33%) < C in C_2H_2 (sp, 50%).

(e) Atomic and ionic radius: χ decreases as the radius increases, $\chi \propto 1/r$.

(f) Effective nuclear charge: χ increases with Z_{eff}, $\chi \propto Z_{\text{eff}}$.

Formula 13.4 Dependence of Electronegativity

$$\chi \propto \frac{1}{\text{atomic size}} \qquad \chi \propto Z_{\text{eff}} \propto \frac{\text{positive charge}}{\text{negative charge}}$$

13.5 Periodic variation

- **In a period:** χ increases left to right (size contracts, Z_{eff} increases), up to the halogens.
- **In a group (representative elements):** χ generally decreases down the group (size increases, Z_{eff} per outer electron almost constant).

Exceptions. (a) In group 18, χ is taken as zero (no bond formation) — Xe and Kr compounds excepted. (b) χ of Cs and Fr is the same (0.7): from Cs to Fr only one shell is added while the nuclear charge rises by +32. (c) In group 12, χ increases down the group (Zn 1.6, Cd 1.7, Hg 1.9) because of the lanthanoid contraction. (d) In group 13, χ *increases* down the group from Al (1.5) to Ga (1.6), In (1.7), Tl (1.8) because of the transition contraction: Ga > Al.

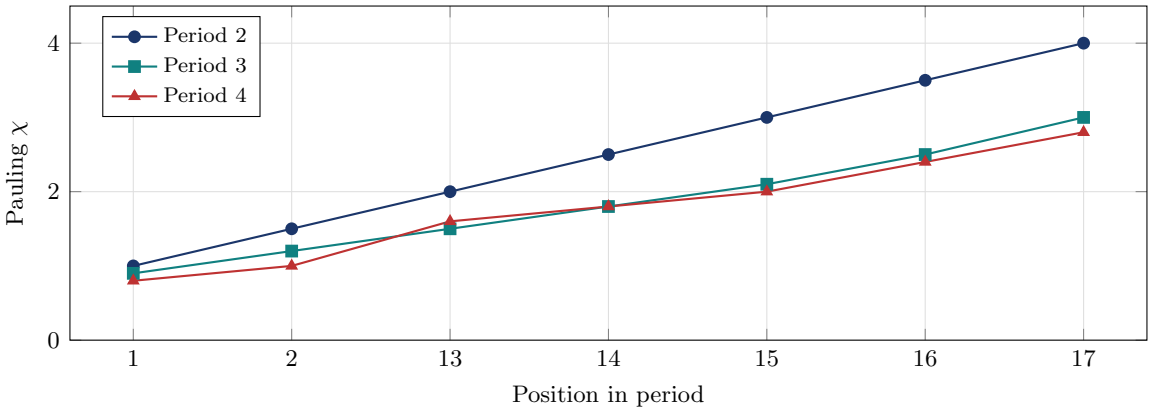


Figure 18. Electronegativity increases across a period and decreases (mostly) down a group.

13.6 Applications of electronegativity

13.6.1 Metallic and non-metallic nature

Low electronegativity ⇒ metals; high electronegativity ⇒ non-metals. Metallic character *increases down a group and decreases along a period.*

13.6.2 Bond length

Electronegativity difference *shortens* a covalent bond (Schomaker–Stevenson), so for a given pair of sizes a larger $\Delta\chi$ gives a shorter bond. In the hydrogen halides the combined effect of the very small F and the large $\Delta\chi$ gives HF the shortest H–X bond.

Formula 13.5 Effect of $\Delta\chi$ on Bond Length

$$\boxed{d_{\mathrm{A-B}} = r_{\mathrm{A}} + r_{\mathrm{B}} - 0.09\,\Delta\chi} \;\Rightarrow\; \Delta\chi\uparrow \Longrightarrow\; d\downarrow$$

Bond	$\Delta\chi$	$r_{\mathrm{H}} + r_{\mathrm{X}}$ / pm	d calculated / pm	d observed / pm
H–F	1.9	108	91	92
H–Cl	0.9	136	128	127
H–Br	0.7	151	145	141
H–I	0.4	170	166	161

13.6.3 Further applications

(a) Acidic strength of hydrides: within a period the acidity increases with the electronegativity of M in M–H: $NH_3 < H_2O < HF$; $SiH_4 < PH_3 < H_2S < HCl$ (and also increases down a group, e.g. HF < HCl < HBr < HI, because the M–H bond weakens).

(b) Nature of the bond (Hannay–Smith, Gallis):

Formula 13.6 Percentage Ionic Character

$$\boxed{\%\ \text{ionic} = 16\,(\chi_{\mathrm{A}} - \chi_{\mathrm{B}}) + 3.5\,(\chi_{\mathrm{A}} - \chi_{\mathrm{B}})^2}$$

Conditions

If $\chi_{\mathrm{A}} - \chi_{\mathrm{B}} \geq 2.1$ then ionic % > 50% (ionic bond); if ≤ 2.1 then < 50% (covalent). By Gallis: $\Delta\chi \geq 1.7$ ionic, ≤ 1.7 covalent. If $\chi_{\mathrm{A}} = \chi_{\mathrm{B}}$ the bond is non-polar (H–H, F–F). For $\chi_{\mathrm{A}} > \chi_{\mathrm{B}}$ with small difference the bond is polar covalent $\mathrm{A}^{\delta-}-\mathrm{B}^{\delta+}$.

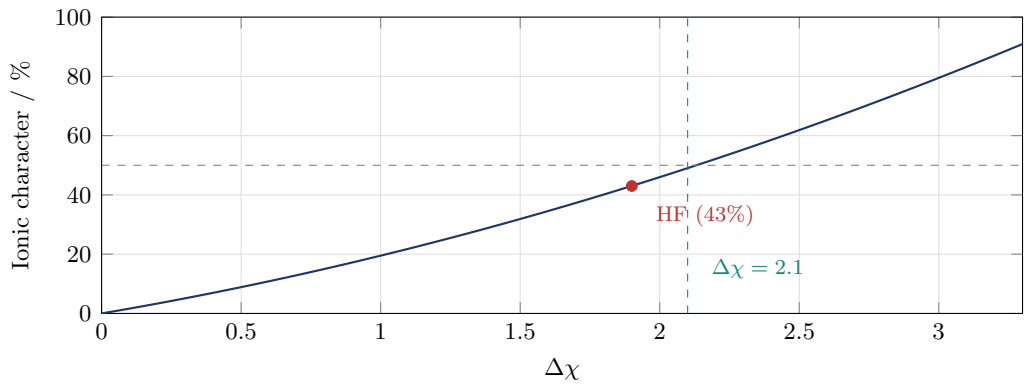


Figure 19. Hannay–Smith relation. For HF, $\Delta\chi = 4.0 - 2.1 = 1.9$: $16(1.9) + 3.5(1.9)^2 = \mathbf{43}\%$.

(c) Nature of hydroxides (Gallis): in A–O–H compare $|\chi_O - \chi_A|$ with $|\chi_O - \chi_H| = 1.4$. If $\chi_A < 1.7$ (metal) the A–O bond breaks (**basic**: $AOH \rightarrow A^+ + OH^-$); if $\chi_A > 1.7$ (non-metal) the O–H bond breaks (**acidic**: $AOH \rightarrow H^+ + AO^-$). NaOH: $\chi_O - \chi_{Na} = 2.6 > 1.4$ basic; ClOH: $\chi_O - \chi_{Cl} = 0.5 < 1.4$ acidic.

(d) Nature of oxides: basic character increases down a group and decreases along a period; acidic character does the reverse and also increases with oxidation number. Left-hand oxides are basic (Na_2O), right-hand ones acidic (Cl_2O_7), middle ones amphoteric (Al_2O_3, ZnO, BeO, SnO, PbO, As_2O_3, Sb_2O_3) or neutral (CO, NO, N_2O, H_2O). Acidic order: $Al_2O_3 < SiO_2 < P_2O_5 < SO_3 < Cl_2O_7$; $N_2O_3 < N_2O_5$; $MnO < Mn_2O_3 < MnO_2 < MnO_3 < Mn_2O_7$.

Worked Example 13.5 Percentage ionic character of HF and acidic strength of hydrides (ALLEN)

(i) $\Delta\chi = 1.9$: $16(1.9) + 3.5(1.9)^2 = 30.4 + 12.6 =$ **43%**. (ii) Acidic character of hydrides increases left to right across a period and down a group, so $SiH_4 < PH_3 < H_2S$. (iii) Basic nature of IA hydroxides increases down the group (larger cation, weaker M–O bond): $CsOH > RbOH > KOH > NaOH > LiOH$. (iv) Acidic order of oxides: $CaO < CO < CO_2 < N_2O_5 < SO_3$.

14 Diagonal Relationship

Elements of the second period resemble the elements of the third period that are placed *diagonally* to their right (Li–Mg, Be–Al, B–Si). The resemblance is attributed to the similar values of **polarising power (charge/size ratio)** of their cations, to similar electronegativities, and to similar sizes.

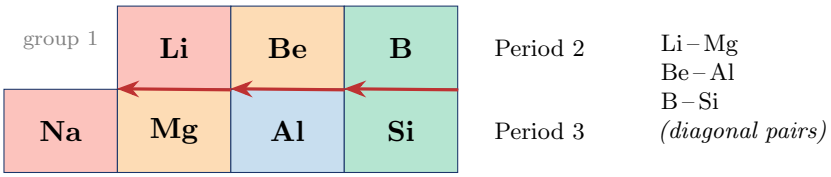


Figure 20. Diagonal relationship: movement to the right (higher charge) and downward (larger size) leaves polarising power nearly unchanged.

Species	Li^+	Mg^{2+}	Be^{2+}	Al^{3+}	B^{3+}	Si^{4+}
Radius / pm (Pauling)	60	65	31	50	20	41
Charge z	1	2	2	3	3	4
Charge/radius (×100 pm)	1.7	3.1	6.5	6.0	15	9.8
Electronegativity of atom	1.0	1.2	1.5	1.5	2.0	1.8

Interpretation

Down a group the charge stays the same but the radius grows, so charge/size falls; across a period the charge rises and the radius falls, so it rises. Moving *diagonally* the two effects partly cancel, so the polarising power and the electronegativity are similar (Be/Al: 1.5 and 1.5). The resemblance is strongest for Be–Al and Li–Mg.

14.1 Lithium resembles magnesium

(a) Li forms covalent compounds like Mg (LiCl is soluble in alcohol and pyridine, and is deliquescent as $MgCl_2$); the rest of the alkali metals form ionic compounds.

(b) Li forms the *normal oxide* Li_2O like Mg on burning in air; the rest form peroxides (Na_2O_2) and superoxides (KO_2).

(c) The carbonate, nitrate and hydroxide of Li decompose on heating to give the oxide (e.g. $Li_2CO_3 \rightarrow$

$Li_2O + CO_2$, like $MgCO_3 \rightarrow MgO + CO_2$), whereas the salts of other alkali metals are thermally stable.

(d) Li and Mg both burn in air to give the oxide *and nitride* (Li_3N, Mg_3N_2); other alkali metals give only the oxide (or higher oxides).

(e) Li_2CO_3, LiF and Li_3PO_4 are sparingly soluble, as $MgCO_3$, MgF_2 and $Mg_3(PO_4)_2$; both form hydrated chlorides ($LiCl{\cdot}2H_2O$, $MgCl_2 \cdot 6H_2O$).

14.2 Beryllium resembles aluminium

(a) Both form covalent compounds.

(b) The oxides and hydroxides of both are *amphoteric* (BeO, Al_2O_3; $Be(OH)_2$, $Al(OH)_3$) and both metals dissolve in NaOH giving H_2 ($[Be(OH)_4]^{2-}$, $[Al(OH)_4]^-$).

(c) Both oxides are very hard, high-melting solids (BeO, corundum Al_2O_3), and both metals are rendered passive by concentrated HNO_3.

(d) The chlorides $BeCl_2$ and $AlCl_3$ are electron-deficient covalent halides that act as *Lewis acids* and are polymeric/dimeric in structure (Al_2Cl_6).

(e) Both form carbides that give methane with water (Be_2C, Al_4C_3).

14.3 Boron resembles silicon

(a) The oxides B_2O_3 and SiO_2 are *acidic* and glass-forming (borosilicate glass).

(b) Both are non-metals (metalloids) with high melting points, and both react with NaOH liberating hydrogen: $Si + 2NaOH + H_2O \rightarrow Na_2SiO_3 + 2H_2$.

(c) Their halides (BCl_3, $SiCl_4$) fume in moist air since on hydrolysis they give HX: $BCl_3 + 3H_2O \rightarrow H_3BO_3 + 3HCl$; $SiCl_4 + 4H_2O \rightarrow Si(OH)_4 + 4HCl$.

(d) Both form volatile, reactive covalent hydrides (boranes and silanes).

Why diagonal relationship fails after the third period

Beyond B and Si the charges and sizes of the ions no longer balance, and the d-block and f-block contractions distort the trends; the concept is mainly useful for the second-period elements Li, Be, B.

Part VI — Exam Tools

15 Master Formula Atlas

Family	Formula	Condition / keyword
Döbereiner	$A_r(\text{mid}) = \dfrac{A_r(1) + A_r(3)}{2}$	triads
Moseley	$\sqrt{\nu} = a(Z - b)$	X-ray spectra
Period length	$N_p = 2\lfloor (p+2)/2 \rfloor^2$	2, 8, 8, 18, 18, 32, 32
Next member	$Z_{\text{below}} = Z_{\text{above}} + N$	magic numbers
Group number	s: n_s; p: $10 + n_s + n_p$; d: $n_d + n_s$	valence configuration
Screening	$Z_{\text{eff}} = Z - \sigma$	Slater's rules
Slater (ns, np)	$\sigma = 0.35N_s + 0.85N_{n-1} + 1.00N_{\le n-2}$	$1s$: 0.30
Slater (nd, nf)	$\sigma = 0.35N_s + 1.00N_{\text{left}}$	right-hand electrons: 0
Valency	n_{val} (groups 1,2,13,14); $8 - n_{\text{val}}$ (15–18)	electronic concept
Covalent radius	$r = d/2$; $d_{AB} = r_A + r_B - 0.09\Delta\chi$	single bond

Family	Formula	Condition / keyword
Metallic radius	fcc $\frac{a}{2\sqrt{2}}$; bcc $\frac{\sqrt{3}a}{4}$; sc $\frac{a}{2}$	metal lattice
Ionic radius	$r_+ = \frac{d(Z_- - \sigma)}{(Z_+ - \sigma) + (Z_- - \sigma)}$	Pauling, isoelectronic pair
Isoelectronic size	$r \propto 1/Z$	same number of electrons
H-like IE	$\mathrm{IE} = 13.6Z^2/n^2$ eV $= 1312Z^2/n^2$ kJ mol^{-1}	one electron
Photon method	$\mathrm{IE} = N_A hc/\lambda_0 = 1.196 \times 10^5/\lambda_0$ (kJ mol^{-1}, nm)	photo-ionisation
Total ionisation	$E = (m/M)\sum \mathrm{IE}_i$	mass sample
IE factors	$\mathrm{IE} \propto Z_{\mathrm{eff}},\ 1/r,\ 1/\sigma$; $s > p > d > f$	half/full-filled exceptions
Oxidation state	$\Delta\mathrm{IE} \geq 16$ eV: stable; ≤ 11 eV: higher state	successive IE
Electron gain	$\mathrm{EA} = -\Delta H_{\mathrm{eg}}$; $\Delta H_{\mathrm{eg}_2} > 0$	isolated gaseous atom
Pauling	$\chi_A - \chi_B = 0.208\sqrt{\Delta}$ (kcal), $0.102\sqrt{\Delta}$ (kJ)	$\Delta = E_{AB} - \sqrt{E_{AA}E_{BB}}$
Mulliken	$\chi_P = \frac{\mathrm{IE} + \mathrm{EA}}{5.6}$	eV
Allred–Rochow	$\chi = 0.359Z_{\mathrm{eff}}/r^2 + 0.744$	r in Å
Ionic character	$16\Delta\chi + 3.5\Delta\chi^2$	$\geq 50\%$ at $\Delta\chi \approx 2.1$
Schomaker–Stevenson	$d_{AB} = r_A + r_B - 0.09\lvert\chi_A - \chi_B\rvert$	bond length (Å)

16 Keyword Decision Tree

Recognise the property from the wording of the question.

- "successive IE", "large jump" → number of valence electrons = number of electrons before the jump.
- "largest/smallest radius", "isoelectronic" → compare Z (higher Z, smaller size); "same group, 5d vs 4d" → lanthanoid contraction.
- "anomalous IE order" → check Be/B, Mg/Al (s penetration); N/O, P/S (half-filled); Al/Ga, In/Tl (d, f shielding).
- "electron affinity" → halogens highest; Cl > F, S > O; noble gases, Be, Mg, N zero or positive; second EA always positive.
- "bond energies given, find χ" → Pauling; "IE and EA given" → Mulliken, divide by 5.6; "Z_{eff} and radius given" → Allred–Rochow.
- "bond length of A–B" → $r_A + r_B - 0.09\Delta\chi$; "edge length a" → metallic radius.
- "which group/period for Z" → configuration, then group formula; "successive element in a group" → magic numbers.
- "acidic/basic oxide or hydroxide" → electronegativity of the central atom (Gallis 1.7 rule), oxidation number.
- "resembles" → diagonal relationship (Li–Mg, Be–Al, B–Si).

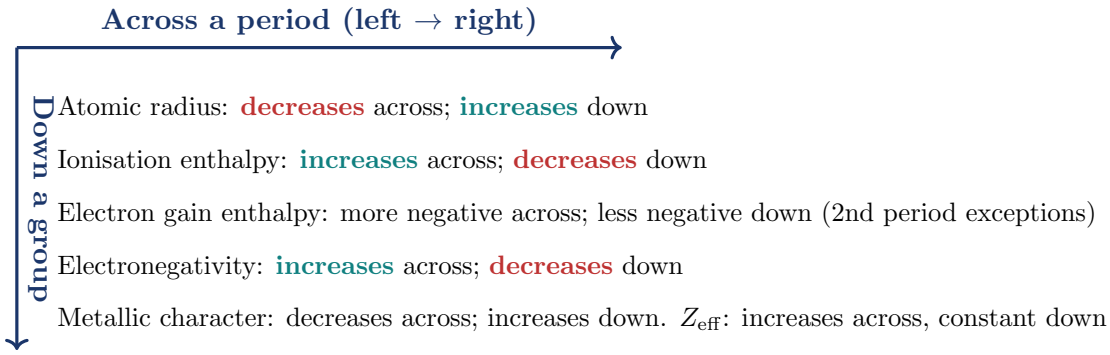


Figure 21. Master chart of periodic trends for representative elements.

17 Symbol and Sign Glossary

Quick reference

Z = atomic number; $Z_{\text{eff}} = Z - \sigma$ = effective nuclear charge; σ = screening constant; n^* = effective principal quantum number; r = radius; χ = electronegativity; IE_k = kth ionisation enthalpy (> 0); ΔH_{eg} = electron gain enthalpy (usually < 0); EA = $-\Delta H_{\text{eg}}$ for the first electron; N_p = elements in period p. 1 eV = 96.485 kJ mol^{-1} = 23.06 kcal mol^{-1}; 1 kcal = 4.184 kJ; 1 Å = 100 pm.

Last-minute exam traps

- IE_2 of Li (7298) is far larger than IE_2 of Be (1757); IE_2 of B (2427) > Be.
- Be, Mg (ns^2) and N, P (np^3) are the exceptions in IE; and in EA the *signs* are opposite to IE trends in some elements — read whether ΔH or EA is asked.
- Electronegativity: F > Cl; electron affinity: Cl > F.
- Radius order for isoelectronic species follows Z (lower Z, bigger); for ions of the same element, more positive means smaller.
- Zn, Cd, Hg are d-block but not transition elements. He is s-block but placed in group 18.
- Atomic radius of noble gases is the van der Waals radius, hence the largest in a period.
- Use the correct unit conversion: 1 eV = 96.5 kJ mol^{-1}; ALLEN's 1 cal = 4.2 J is approximate.

Source notes and corrections

While extracting the ALLEN notes the following points were found to be inconsistent or out of date and have been corrected in this book:

Statement in the source	Treatment here
Period 7 incomplete, 26 elements	Complete with 32 elements (elements 113–118 named)
s-block "12 elements"; p-block "31 (excluding He)"	12 s-block elements excluding H, He; p-block has 36 elements for $Z \le 118$ (older counts are 30/31)
Cs and Fr "liquid"	Solids at 298 K (melting just above room temperature)
f-block general formula with complete inner shells	$(n-2)f^{1-14}(n-1)d^{0-1}ns^2$
Cr $Z_{\text{eff}} = 3.45$	Actual $3d^5 4s^1$ gives 2.95
χ of C from data = 2.5	Exact calculation gives 2.60 (literature 2.55)
IP of Tl, Pb "group IVA"	Group 13 (Tl) and 14 (Pb)
Electron-gain table with B −83, N +31	NCERT values used; pattern unchanged
Mulliken value 3.03 for Cl	17/5.6 = 3.04

End of the Periodic Table and Periodicity atlas. Every boxed result is traceable to a derivation or justification given above.